

Convergent Evidential Protocol (CEP) v1.4

First Demonstration: The Shroud of Turin

A Structured, Quantitative Framework for Evaluating Complex, Heterogeneous Evidence Under Uncertainty

Author: Tyler Leroux

Date: June 2026

Version: 1.4

Version History

Version	Major Changes
1.0	Initial convergence framework with evidence tiers, hypothesis prediction tables, and domain structure.
1.1	Added calibration confidence flags, domain dependency modeling, Monte Carlo probabilistic synthesis, sensitivity analysis (including Skeptical Vector), parsimony assessment, and the Operational Falsification and Updating Protocol.
1.2	Refined framing for consistency with Domain 7; strengthened positioning of the CEP as a generalizable methodology; added Version History and Methodological Significance sections.
1.3	Introduced explicit weighted criterion aggregation. Domain-level likelihood ranges are now mathematically derived from individual criteria using pre-defined Tier and Weight Category scores. This change makes the input layer fully auditable, modular, and traceable to specific evidence characteristics. Updated statistical deliverables to reflect the new bottom-up aggregation.
1.4	Introduced adjustable priors (default flat) and Beta distribution sampling in the Monte Carlo synthesis engine for more realistic uncertainty modeling. Retained the 3-layer domain weighting system. Updated all posterior summaries, Bayes factors, and derived domain likelihood ranges to reflect the refined sampling methodology.

Methodological Significance

This document presents the **Convergent Evidential Protocol (CEP)**, a structured, quantitative methodology for evaluating complex, heterogeneous evidence under conditions of uncertainty and interpretive disagreement.

While developed and first demonstrated in the context of the Shroud of Turin, the protocol is designed to be generalizable. The same core mechanisms can be adapted to other complex evidential problems in history, science, and anomalous phenomena. It provides a transparent and auditable framework for integrating evidence across multiple disciplines, explicitly modeling dependencies between lines of evidence, propagating uncertainty through Monte Carlo simulation, testing robustness under adversarial assumptions, and specifying in advance how new data would strengthen, weaken, or falsify competing hypotheses.

The CEP combines several methodological components into a single coherent system: explicit evidence tiering and weighting, criterion-level likelihood estimation with weighted aggregation into domain composites, correlation-adjusted Monte Carlo synthesis, multi-scenario sensitivity testing (including adversarial “Skeptical Vector” analysis), parsimony assessment across hypotheses, and a pre-specified Operational Falsification and Updating Protocol.

The Shroud of Turin was selected as the first full-scale demonstration because it presents an unusually demanding test case. It requires the rigorous integration of physical, chemical, forensic, historical, and probabilistic evidence within a single evaluative structure, while navigating long-standing tensions between domains.

Foundational Elements (Locked)

This analysis evaluates four competing hypotheses for the origin and image-forming mechanism of the Shroud of Turin:

- **H_A** — Medieval Artifact
- **H_B** — Natural Ancient Mechanism
- **H_C** — Unknown Physical Mechanism
- **H_D** — Resurrection-linked Mechanism (with physical and historical constraints)

Evidence is assessed domain by domain using explicit likelihood ranges $P(E \mid H)$, which are aggregated via Monte Carlo simulation in Domain 7.

The structure is intentionally locked at the criterion level in the current version to maintain analytical integrity and permit transparent, auditable updating as new data emerge, while preserving the full evidential tension between physical/forensic convergence and radiocarbon/historical constraints.

A note on hypothesis framing is warranted at the outset. **H_D** (Resurrection-linked) is not treated as an unconstrained miracle hypothesis. The model imposes real constraints on H_D by requiring it to align with the physical, forensic, and historical evidence. This prevents it from functioning as an explanatory wildcard capable of accommodating any observation.

1. Evidence Quality Matrix

Every criterion in the Master Criterion Registry is assigned two independent attributes that together determine its influence on the domain-level composites:

- **Evidence Tier (A–E)**: Reflects the intrinsic strength and directness of the underlying data or reasoning.
- **Weight Category (High / Medium-High / Medium / Low)**: Reflects the criterion's relative importance to the overall hypothesis evaluation within its domain.

These two attributes are combined mathematically (see 1.1 below) to produce the domain-level likelihood ranges that feed the Monte Carlo synthesis. The classification of every criterion is locked in the current version to maintain full traceability and reproducibility.

Tier	Description	Characteristics	Examples
A	Direct physical measurements	Objective, instrument-based, reproducible data	Fiber microscopy, 3D encoding measurements, radiocarbon results
B	Laboratory replications	Controlled experimental reproduction of observed features	ENEA VUV experiments, modern image reproduction attempts

C	Historical documentation	Contemporary records, chain of custody, eyewitness or documentary sources	First historical appearance, Byzantine references, ownership records
D	Expert testimony / consensus	Specialist analysis or broad agreement among qualified experts	Anatomical assessments by forensic pathologists, textile expert opinions
E	Interpretive inference	Reasonable but non-direct conclusions drawn from data	Inferences about image formation mechanism from combined features

1.1 Weighted Aggregation Scoring Rubric

Domain-level likelihood ranges are not assigned holistically. They are derived bottom-up from the individual criteria through explicit weighted aggregation. Each criterion receives a combined weight equal to its Evidence Tier score multiplied by its Weight Category score. These scores function as relative multipliers that give greater influence to higher-quality and higher-impact evidence while ensuring every criterion still contributes to the composite.

The composite range for each hypothesis within a domain is calculated as the weighted average of the criterion-level bounds:

$$\text{Composite Range} = \frac{\sum(\text{Criterion Range} \times \text{Tier Score} \times \text{Weight Score})}{\sum(\text{Tier Score} \times \text{Weight Score})}$$

The scoring values used throughout this analysis (and implemented directly in

`shroud_mc_v1.4_weighted.py`) are:

Evidence Tier	Score	Weight Category	Score
A – Direct physical measurements	1.00	High	1.00
B – Laboratory replications	0.85	Medium-High	0.80
C – Historical documentation	0.70	Medium	0.60
D – Expert testimony / consensus	0.55	Low	0.40
E – Interpretive inference	0.40		

Implementation Note

These combined weights are applied uniformly across all six domains. Resulting domain composites are fully auditable: every value in the

`aggregated_likelihood_ranges` output can be traced back to specific criteria, tiers, and weight categories. Alternative scoring schemes can be substituted for sensitivity testing without altering the rest of the framework.

1.2 Three-Layer Domain Weighting System

Domain-level influence in the Monte Carlo synthesis is determined through a transparent, multi-factor weighting process rather than by subjective judgment. Each domain is assigned a derived weight based on three independent layer scores that are combined mathematically:

- **Evidence Strength** — Reflects the overall quality and evidential weight of the criteria contained within the domain. Domains built from higher-tier, higher-impact criteria receive stronger influence.
- **Diagnosticity** — Measures how effectively the domain differentiates between the four hypotheses. This is calculated from the spread of likelihood midpoints across H_A through H_D. Domains that create clearer separation between hypotheses contribute more to the synthesis.
- **Independence** — Accounts for correlation between domains. Domains that are less redundant with other lines of evidence receive higher weight, reducing the risk of double-counting related information.

These three normalized layer scores are combined using the following weighted equation:

$$W_d = (S_{ES} \times w_{ES}) + (S_{Diag} \times w_{Diag}) + (S_{Ind} \times w_{Ind})$$

Where S_{ES} , S_{Diag} , and S_{Ind} are the normalized scores for Evidence Strength, Diagnosticity, and Independence for domain d , and w_{ES} , w_{Diag} , and w_{Ind} are the relative layer weights. These layer weights were derived through pairwise comparisons that prioritize Evidence Strength most heavily, followed by Diagnosticity, with Independence receiving the smallest share. This ordering reflects the principle that high-quality

evidence should carry the greatest influence, provided it also meaningfully distinguishes between hypotheses and does not simply duplicate information already captured elsewhere.

The resulting domain weights are then scaled so that their mean equals 1.0. This scaling is applied for interpretability: a weight of 1.0 represents average influence across domains, while values above or below 1.0 indicate stronger or weaker influence relative to the average. The scaling does not alter the relative differences between domains or the final Monte Carlo results.

This three-layer approach ensures that domains with stronger, more distinctive, and more independent evidence exert greater pull on the final posteriors, while still allowing every domain to contribute proportionally. The full layer scores and derived domain weights for the current analysis are reported alongside the Monte Carlo results in Domain 7.

2. Master Criterion Registry

The following seven domains and their criteria form the current analytical structure of the framework. This framework is intentionally stable to support rigorous and reproducible analysis.

New criteria may be added in future versions when justified by substantial new evidence, provided they are properly integrated into the existing domains and Monte Carlo correlation model, with clear documentation of any resulting changes.

Domain 1: Image Formation Physics

- 1.1 Extreme superficiality (fiber depth)
- 1.2 Absence of pigments and directionality
- 1.3 3D spatial encoding
- 1.4 Lack of scorching / thermal damage
- 1.5 Laboratory replication analogues (ENEA and related work)

Domain 2: Radiocarbon Dating & Sampling Integrity

- 2.1 1988 radiocarbon results and inter-laboratory consistency
- 2.2 Sample location and selection concerns

- 2.3 Reweaving hypothesis
- 2.4 Statistical heterogeneity and contamination hypotheses
- 2.5 Lack of modern independent retesting

Domain 3: Forensic / Medical Coherence

- 3.1 Anatomical accuracy of wounds and body positioning
- 3.2 Blood flow consistency and directionality
- 3.3 Roman scourging pattern match
- 3.4 Wrist vs. palm nailing placement
- 3.5 Side wound characteristics
- 3.6 Post-mortem indicators

Domain 4: Comparative Artifact Analysis

- 4.1 Comparison with known medieval relics and image techniques
- 4.2 Statistical uniqueness of the Shroud image properties
- 4.3 Success/failure of modern reproduction attempts

Domain 5: Historical Provenance

- 5.1 First undisputed historical appearance (14th century)
- 5.2 Possible earlier links (Edessa / Constantinople)
- 5.3 Documentary and iconographic continuity
- 5.4 Historical gaps

Domain 6: Investigation History

- 6.1 Access restrictions and testing decisions
- 6.2 Sampling history and methodology documentation
- 6.3 Publication patterns and treatment of anomalous findings

Domain 7: Explanatory Power (Synthesis Layer)

This domain performs the integrative work:

- Domain-level composite likelihood construction via weighted aggregation of individual criteria (see §1.1 and §1.2)
- Monte Carlo probabilistic aggregation using 8,000 iterations with correlated draws via Gaussian copula and Beta distribution sampling
- Adjustable priors with sensitivity testing

- Sensitivity analysis (base case, Skeptical Vector, and Hyper-Skeptical Audit)
 - Parsimony / complexity assessment across hypotheses
 - Operational Falsification and Updating Protocol
 - Final Bayesian synthesis, robustness summary, and ranked comparative evaluation
-

Domain 1: Image Formation Physics

Domain Overview

Domain 1 evaluates the physical characteristics of how the Shroud image was formed. It focuses on five high-weight criteria related to the image's microscopic properties, spatial encoding, and experimental replicability. This domain contains some of the strongest empirical constraints in the entire evaluation.

Criterion 1.1: Extreme Superficiality (Fiber Depth)

Evidence E

The body image on the Shroud of Turin is confined exclusively to the outermost 0.2–0.6 μm layer of individual linen fibers (the primary cell wall). The internal medulla of the fibers remains completely colorless and unaffected. This extreme superficiality has been confirmed through multiple independent microscopic examinations, including optical and scanning electron microscopy plus fiber cross-section analyses performed by STURP researchers (Heller & Adler, 1981) and subsequent textile chemists. The coloration is uniform across both the frontal and dorsal images and shows no evidence of deeper penetration or subsurface diffusion.

Strongest Alternative Counter

The image was created by a medieval artistic or contact-chemical process (such as a Maillard-type reaction or low-energy vaporographic method). Laboratory versions of such processes can produce superficial discoloration on linen without deep fiber penetration. However, no known or experimentally demonstrated medieval-era technique has achieved this effect consistently across the entire frontal and dorsal body images at the observed scale and uniformity while avoiding residues, directionality, or deeper penetration.

Contradiction Handling

The 1988 radiocarbon dating creates significant tension for any hypothesis requiring a 1st-century origin. This tension is carried forward and treated as a factor that reduces support for H_B, H_C, and H_D until addressed by other criteria. At the same time, the extreme superficiality stands as an independent physical constraint on possible image-formation mechanisms regardless of the cloth's age.

Expert Consensus Tracking

The extreme superficiality of the image (limited to the primary cell wall) is broadly accepted among researchers who have directly examined the Shroud fibers under microscopy.

Evidence Quality Tier & Weight Category

Evidence Tier: A (Direct physical measurements)

Weight Category: High

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.10 - 0.20$$

Justification: While some low-energy chemical processes can produce superficial discoloration, no documented medieval technique (artistic, contact-printing, or chemical) has been shown to reliably restrict coloration so precisely to the outermost 0.2–0.6 μm across the entire frontal and dorsal images while avoiding deeper penetration, residues, or directional artifacts. Experimental attempts at Maillard-type or vaporographic methods have not yet scaled successfully to this level of uniformity and precision.

$$P(E | H_B) \approx 0.15 - 0.28$$

Justification: Natural post-mortem processes in an ancient burial setting could theoretically produce some superficial effects. However, achieving such consistent and extreme depth limitation across the full body image without additional assumptions remains difficult to explain under known natural mechanisms.

$$P(E | H_C) \approx 0.40 - 0.60$$

Justification: Laboratory work has demonstrated that certain high-energy, short-duration processes can produce coloration limited to the primary cell wall. This raises

the likelihood under an unknown physical mechanism, though no established natural context has yet been shown to produce this result at the observed scale and uniformity.

$$P(E | H_D) \approx 0.55 - 0.72$$

Justification: This hypothesis permits a directed process capable of precise, surface-limited interaction with the linen fibers. The observed extreme superficiality is consistent with the level of control this category allows.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 4.8 \mid H_D \text{ vs } H_B \approx 3.2 \mid H_D \text{ vs } H_C \approx 1.5$$

Interpretation

Criterion 1.1 provides one of the strongest physical constraints on possible image-formation mechanisms. The extreme superficiality significantly lowers the likelihood under H_A and moderately lowers it under H_B , while remaining compatible with both H_C and H_D . This criterion carries high weight due to its direct physical nature (Tier A) and will exert meaningful downward pressure on conventional medieval artifact hypotheses going forward.

Criterion 1.2: Absence of Pigments and Directionality

Evidence E

Microscopic and chemical analysis of the Shroud image shows a complete absence of pigments, binders, or particulate matter. There is also no evidence of brush strokes, rubbing, or any directional application method. The coloration appears to result from a chemical change in the linen fibers themselves rather than the addition of any foreign substance. This has been confirmed through visible-light and UV-fluorescence microscopy, X-ray fluorescence, Raman spectroscopy, and pyrolysis-mass spectrometry performed by STURP researchers and subsequent independent analysts.

Strongest Alternative Counter

The image was created using a medieval artistic technique such as dry brushing, powder rubbing, or a low-energy chemical contact method that did not require traditional pigments or binders. Some experimental attempts have produced superficial discoloration without obvious directional marks. However, achieving a complete

absence of any particulate residue or binder across the entire frontal and dorsal images at the observed scale and uniformity has not been demonstrated with known medieval-era techniques.

Contradiction Handling

While some artistic or contact methods can reduce visible directionality, reproducing the complete absence of any particulate material or binder across the entire image remains difficult to achieve consistently with known medieval techniques. This tension is noted but does not eliminate H_A as a possibility. The absence of pigments and directionality functions as an independent physical constraint on image-formation mechanisms.

Expert Consensus Tracking

The absence of pigments, binders, and directional application is broadly accepted among researchers who have performed direct microscopic and spectroscopic examination of the Shroud.

Evidence Quality Tier & Weight Category

Evidence Tier: A (Direct physical measurements)

Weight Category: High

Likelihood Estimates $P(E | H)$

$P(E | H_A) \approx 0.08 - 0.18$

Justification: Most medieval artistic methods (painting, rubbing, or printing) leave behind some trace of pigment, binder, or directional application. While some experimental techniques have reduced these traces, none have fully eliminated them across the scale and uniformity seen on the Shroud — particularly the complete absence of particulate matter or binder residues over both the frontal and dorsal images.

$P(E | H_B) \approx 0.20 - 0.35$

Justification: Certain natural chemical processes in a burial context could theoretically alter the linen fibers without introducing external pigments. However, achieving completely uniform results without any directional bias or residue across both frontal and dorsal images requires additional assumptions.

$$P(E | H_C) \approx 0.45 - 0.65$$

Justification: An unknown physical or radiative process could produce fiber coloration through direct chemical alteration without adding pigments or creating directionality. This aligns reasonably well with the observed characteristics.

$$P(E | H_D) \approx 0.58 - 0.75$$

Justification: A directed process could produce precise chemical modification of the fibers without the addition of any external material or directional artifacts, which is consistent with the complete absence of pigments and directionality.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 5.5 \mid H_D \text{ vs } H_B \approx 2.8 \mid H_D \text{ vs } H_C \approx 1.4$$

Interpretation

Criterion 1.2 adds significant constraint against conventional artistic explanations. The total absence of pigments and directional marks substantially lowers the likelihood under H_A while remaining compatible with both H_C and H_D . This criterion carries high weight (Tier A) and further weakens simple medieval artifact hypotheses.

Criterion 1.3: 3D Spatial Encoding

Evidence E

The Shroud image contains accurate three-dimensional spatial information. When processed through the VP-8 Image Analyzer and subsequent digital photometric methods, the intensity of the image correlates with the distance between the cloth and the body surface. Areas where the cloth would have been closer to the body appear brighter, producing a recognizable three-dimensional relief. This distance-to-intensity encoding is present across both the frontal and dorsal images and has been confirmed through multiple independent analyses of the raw image data (Jackson, Jumper, and subsequent photometric/3D reconstruction studies).

Strongest Alternative Counter

The 3D effect is an artifact of the image processing method rather than an inherent property of the original image. Medieval contact techniques (such as rubbing or bas-relief methods) could produce shading variations that, when processed through

modern imaging software, create the appearance of 3D information. However, multiple independent analyses have shown that the distance-to-intensity correlation exists in the raw image data itself and is not solely generated by the VP-8 or later processing.

Contradiction Handling

While image processing can enhance contrast, multiple independent analyses have shown that the distance-to-intensity relationship exists in the raw data and is not solely an artifact of the VP-8 or later software. This reduces but does not eliminate the strength of the counter-argument under H_A. The 3D encoding functions as an independent physical constraint on possible image-formation mechanisms.

Expert Consensus Tracking

The presence of distance-encoded information in the Shroud image is accepted by most researchers who have analyzed the image using photometric and 3D reconstruction methods, though interpretations of how it was formed remain disputed.

Evidence Quality Tier & Weight Category

Evidence Tier: A (Direct physical measurements)

Weight Category: High

Likelihood Estimates $P(E | H)$

$P(E | H_A) \approx 0.06 - 0.15$

Justification: Conventional medieval artistic or contact methods generally do not produce images with accurate, non-contact distance encoding. Creating consistent 3D information that corresponds to actual body-cloth distance would require an unusually sophisticated technique that has not been demonstrated in known medieval artifacts or experimental reproductions.

$P(E | H_B) \approx 0.12 - 0.25$

Justification: Natural processes in a burial setting could theoretically produce some variation in image intensity, but achieving precise, accurate distance encoding across the entire body is difficult to explain under currently understood natural mechanisms without additional assumptions.

$$P(E | H_C) \approx 0.48 - 0.68$$

Justification: Certain high-energy or radiative processes could interact with the cloth in a distance-dependent manner, producing intensity variations that correspond to cloth-to-body distance. This is physically plausible and aligns with some laboratory observations of radiation-based image formation.

$$P(E | H_D) \approx 0.62 - 0.78$$

Justification: A directed process could produce precise, distance-sensitive interaction with the linen, resulting in accurate three-dimensional encoding without physical contact. This is consistent with the level of spatial control permitted under this hypothesis.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 7.2 \mid H_D \text{ vs } H_B \approx 4.1 \mid H_D \text{ vs } H_C \approx 1.5$$

Interpretation

Criterion 1.3 represents one of the most constraining features of the image. The presence of accurate, non-contact 3D spatial encoding significantly weakens H_A and moderately weakens H_B . It remains compatible with both H_C and H_D , with a modest advantage to H_D due to the precision of the encoding. This criterion carries high weight (Tier A) and will play an important role in the overall domain assessment.

Criterion 1.4: Lack of Scorching / Thermal Damage

Evidence E

Despite the presence of visible image coloration on the linen fibers, there is no evidence of scorching, charring, or thermal degradation. Microscopic examination shows that the fibers retain their structural integrity with no signs of heat damage, brittleness, or discoloration consistent with burning. This holds true even in areas of higher image intensity. The absence of thermal effects has been confirmed through optical and scanning electron microscopy, UV fluorescence, and spectroscopic analysis (including FTIR and Raman) performed by STURP researchers and subsequent textile chemists.

Strongest Alternative Counter

The image was formed through a low-temperature or controlled thermal process, or through a non-thermal chemical reaction (such as a Maillard-type reaction) that can produce color without reaching scorching temperatures. Some experimental reproductions have achieved superficial discoloration with minimal or no visible scorching. However, consistently achieving the specific combination of superficial coloration without any detectable thermal damage or structural change across the entire frontal and dorsal images has proven difficult in experimental attempts.

Contradiction Handling

While certain low-energy thermal or chemical methods can avoid obvious charring, reproducing the specific combination of superficial coloration without any thermal damage across the entire image has proven difficult in experimental attempts. This tension is acknowledged and reduces the strength of purely thermal explanations under H_A.

Expert Consensus Tracking

The absence of scorching or thermal damage to the fibers is broadly accepted among researchers who have conducted microscopic and spectroscopic analysis of the Shroud.

Evidence Quality Tier & Weight Category

Evidence Tier: A (Direct physical measurements)

Weight Category: High

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.12 - 0.22$$

Justification: Many proposed medieval formation methods involving heat, scorching, or high-temperature contact tend to produce some degree of thermal damage or charring. While some low-energy techniques avoid this, consistently achieving coloration without any thermal effects across both sides of the cloth remains challenging to demonstrate.

$$P(E | H_B) \approx 0.25 - 0.40$$

Justification: Natural chemical processes in a burial environment could potentially produce fiber discoloration without thermal damage. However, the uniformity and

specificity of the image without any associated heat effects require additional assumptions under known natural mechanisms.

$P(E | H_C) \approx 0.50 - 0.70$

Justification: Certain high-energy but extremely brief processes (such as short-pulse radiation) can produce coloration while depositing minimal heat, avoiding scorching. This is supported by laboratory replication attempts and makes H_C more compatible with the observed lack of thermal damage.

$P(E | H_D) \approx 0.60 - 0.78$

Justification: A precisely controlled process could interact with the fibers in a way that produces coloration without generating sufficient heat to cause scorching or structural damage. This aligns well with the characteristics permitted under this hypothesis.

Bayes Factors (midpoint estimates)

H_D vs $H_A \approx 4.3$ | H_D vs $H_B \approx 2.4$ | H_D vs $H_C \approx 1.3$

Interpretation

Criterion 1.4 further constrains thermal-based explanations. The complete absence of scorching significantly lowers the likelihood under H_A and provides moderate support for both H_C and H_D . This criterion carries high weight (Tier A) and strengthens the case against conventional heat or scorching mechanisms while remaining compatible with non-thermal or precisely controlled processes.

Criterion 1.5: Laboratory Replication Analogues

Evidence E

Laboratory experiments conducted by the ENEA team (Paolo Di Lazzaro et al.) demonstrated that short, intense pulses of vacuum ultraviolet (VUV) radiation can produce superficial coloration on raw linen fibers. The resulting color is limited to the outermost layers of the fibers, shows no scorching, and exhibits some similarities to the Shroud image in terms of depth and appearance. These experiments required precise control of energy delivery and pulse duration within a narrow parameter range. The superficial, non-scorching nature of the coloration has been confirmed through optical and scanning electron microscopy plus spectroscopic analysis of the treated fibers.

Strongest Alternative Counter

The laboratory results are not relevant to the historical formation of the Shroud image. The conditions required (extreme instantaneous energy delivery via VUV) have no plausible medieval or natural ancient counterpart, and the experiments only achieved partial similarity rather than a full reproduction of all Shroud image characteristics — specifically the combination of extreme superficiality, accurate non-contact 3D encoding, and complete absence of pigments or directionality across both frontal and dorsal images at the observed scale and uniformity.

Contradiction Handling

While the ENEA work demonstrates that a physical process *can* produce superficial, non-scorching coloration, it does not explain how such conditions could have occurred in a 1st-century or medieval context. This limitation is acknowledged and prevents the results from strongly favoring any single hypothesis on their own. Additionally, other experimental approaches (such as Maillard-type reactions, vaporographic methods, and chemically treated cloths) have been considered, but none have yet demonstrated the full combination of extreme superficiality, accurate non-contact 3D encoding, and complete absence of pigments or directionality across both frontal and dorsal images at the scale and uniformity observed on the Shroud. The laboratory replication results function as an independent experimental constraint on possible image-formation mechanisms.

Expert Consensus Tracking

The ENEA VUV experiments are widely acknowledged as the most successful laboratory attempts to date at reproducing key physical characteristics of the Shroud image (superficiality and lack of scorching). However, their relevance to the actual historical formation remains debated.

Evidence Quality Tier & Weight Category

Evidence Tier: B (Laboratory replications)

Weight Category: Medium-High

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.05 - 0.12$$

Justification: Medieval technology had no known method capable of generating the type of controlled, high-energy VUV radiation used in the experiments. The laboratory results do not support conventional artistic or low-energy chemical explanations. While some experimental work on Maillard-type reactions and chemically treated cloths has produced superficial discoloration, none have replicated the combination of extreme superficiality, accurate non-contact 3D encoding, and complete absence of pigments or directionality at the observed scale and uniformity.

$$P(E | H_B) \approx 0.10 - 0.20$$

Justification: No known natural burial process has been shown to produce the specific combination of effects demonstrated in the VUV experiments. The results provide limited support for purely natural ancient mechanisms, as they require conditions (high-energy, short-duration radiation) that have no established counterpart in known post-mortem environments.

$$P(E | H_C) \approx 0.55 - 0.75$$

Justification: The successful laboratory replication using a physical radiative process significantly increases the plausibility of an unknown physical mechanism. It demonstrates that at least one non-artistic, non-contact physical process can produce the required superficial characteristics without scorching, although the experiments only achieved partial replication and do not yet explain how such conditions could have occurred historically.

$$P(E | H_D) \approx 0.50 - 0.70$$

Justification: A directed process could theoretically employ or mimic the type of precise, high-energy interaction demonstrated in the laboratory. The results are compatible with this hypothesis, though they do not specifically confirm it and remain limited by the partial nature of the replication.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 8.5 \mid H_D \text{ vs } H_B \approx 4.8 \mid H_C \text{ vs } H_A \approx 7.8 \mid H_C \text{ vs } H_B \approx 4.2$$

Interpretation

Criterion 1.5 provides important experimental evidence that a physical, non-contact process can produce key features of the Shroud image (superficiality without

scorching). This significantly weakens H_A and provides moderate-to-strong support for both H_C and H_D. Because the experiments only achieve partial replication and do not explain the historical mechanism, the criterion is rated Medium-High in weight (Tier B). It strengthens the physical plausibility of non-artistic formation pathways while highlighting the current gap between laboratory results and any fully explanatory historical process.

Domain 1 Summary: Image Formation Physics

Domain Overview

Domain 1 evaluates the physical characteristics of how the Shroud image was formed. It focuses on five high-weight criteria related to the image's microscopic properties, spatial encoding, and experimental replicability. This domain contains some of the strongest empirical constraints in the entire evaluation.

Criteria Summary

Criterion	Description	Weight	Evidence Tier	Primary Impact
1.1	Extreme superficiality (fiber depth)	High	A	Strongly weakens H_A
1.2	Absence of pigments and directionality	High	A	Strongly weakens H_A
1.3	3D spatial encoding	High	A	Strongly weakens H_A
1.4	Lack of scorching / thermal damage	High	A	Weakens H_A, supports H_C & H_D
1.5	Laboratory replication analogues	Medium-High	B	Supports H_C & H_D

Hypothesis Performance in Domain 1

H_A (Medieval Artifact)

This hypothesis performed poorly across Domain 1.

Criteria 1.1, 1.2, and 1.3 significantly lower its likelihood due to the difficulty of explaining extreme superficiality, lack of pigments/directionality, and accurate 3D encoding using known medieval techniques. Criterion 1.4 and 1.5 further reduce plausibility, as thermal or artistic methods struggle to avoid scorching while matching laboratory replication results.

Overall Domain Impact: Substantial weakening of H_A.

H_B (Natural Ancient Mechanism)

This hypothesis performed moderately but faced consistent limitations.

Natural burial processes can account for some superficial effects (Criteria 1.2 and 1.4), but struggle with the precision of 3D encoding (1.3) and the specific results shown in laboratory replication (1.5).

Overall Domain Impact: Moderate weakening. H_B remains viable but is not strongly supported by the physical data in this domain.

H_C (Unknown Physical Mechanism)

This hypothesis performed well.

The combination of extreme superficiality, lack of scorching, 3D encoding, and successful laboratory replication using radiative methods aligns reasonably well with an unknown physical process. Criterion 1.5 provides the strongest support for H_C within this domain.

Overall Domain Impact: Moderate strengthening.

H_D (Resurrection-linked Mechanism)

This hypothesis performed strongest in Domain 1.

The high level of precision, constraint, and non-destructive encoding observed across multiple criteria is most compatible with a directed process. While not proven by this domain alone, the physical characteristics are more easily accommodated under H_D than under H_A or H_B.

Overall Domain Impact: Moderate strengthening.

Key Findings from Domain 1

The physical properties of the image create significant difficulties for conventional medieval artistic explanations (H_A). Natural ancient mechanisms (H_B) are possible

but face challenges explaining the full combination of features, especially 3D encoding and laboratory replication results. Both H_C and H_D remain viable, with H_D showing a slight edge due to the high degree of precision and control implied by the data. Domain 1 establishes strong empirical constraints that later domains (particularly Radiocarbon) must be measured against.

Implications

Domain 1 shifts the comparative standing by significantly reducing the plausibility of H_A while keeping H_C and H_D as the leading contenders from a physical standpoint. These findings will be carried forward and combined with Domain 2 (Radiocarbon), which is expected to exert counter-pressure.

Domain 1 Criterion Registry

This table presents the complete, auditable input layer for Domain 1. These values are used directly by the weighted aggregation function to produce the domain-level likelihood ranges.

ID	Criterion	Tier	Weight	H_A Range	H_B Range	H_C Range	H_D Range
1.1	Extreme Superficiality (fiber depth)	A	High	0.10 – 0.20	0.15 – 0.28	0.40 – 0.60	0.55 – 0.72
1.2	Absence of Pigments and Directionality	A	High	0.08 – 0.18	0.20 – 0.35	0.45 – 0.65	0.58 – 0.75
1.3	3D Spatial Encoding	A	High	0.06 – 0.15	0.12 – 0.25	0.48 – 0.68	0.62 – 0.78
1.4	Lack of Scorching / Thermal Damage	A	High	0.12 – 0.22	0.25 – 0.40	0.50 – 0.70	0.60 – 0.78
1.5	Laboratory Replication	B	Medium-High	0.05 – 0.12	0.10 – 0.20	0.55 – 0.75	0.50 – 0.70

	Analogues						
--	-----------	--	--	--	--	--	--

Derived Domain 1 Likelihood Ranges (Weighted Aggregation)

Hypothesis	Derived Range	Notes
H_A (Medieval Artifact)	0.084 – 0.178	Strongly constrained by criteria 1.1–1.3
H_B (Natural Ancient)	0.168 – 0.303	Moderate support from some natural-process criteria
H_C (Unknown Physical)	0.471 – 0.671	Strong compatibility with physical data
H_D (Resurrection-linked)	0.575 – 0.749	Highest compatibility with precision and control requirements

These derived ranges feed directly into the Monte Carlo synthesis in Domain 7.

Domain 2: Radiocarbon Dating & Sampling Integrity

Domain Overview

Domain 2 examines the 1988 radiocarbon dating results and the surrounding questions about sample integrity. This is the strongest skeptical domain in this evaluation and directly challenges hypotheses requiring a 1st-century origin.

Criterion 2.1: 1988 Radiocarbon Results and Inter-Laboratory Consistency

Evidence E

In 1988, three independent laboratories (Arizona, Oxford, and Zurich) performed radiocarbon dating on samples taken from the Shroud using accelerator mass spectrometry (AMS). All three laboratories produced results that clustered within a relatively narrow range, dating the tested samples to between approximately 1260 and 1390 AD (with 95% confidence). The results were published in *Nature* (Damon et al., 1989) and showed good internal consistency between the three labs.

Strongest Alternative Counter

The 1988 dating accurately reflects the age of the cloth. The Shroud is a medieval artifact produced in the 13th or 14th century, consistent with its first undisputed historical appearance in France. This conclusion assumes that the dated samples are representative of the main body of the cloth.

Contradiction Handling

Significant concerns have been raised about the location from which the samples were taken (the corner area). Critics argue this section may contain medieval repairs or reweaving, meaning the dated material may not represent the original cloth. These concerns are treated as a serious challenge to the representativeness of the results rather than a dismissal of the laboratory measurements themselves. The laboratory measurements themselves function as an independent chronological constraint that must be weighed against the physical and forensic evidence in other domains.

Expert Consensus Tracking

The laboratory measurements and statistical consistency of the 1988 dating are accepted as technically sound by the scientific community. However, the question of whether the samples are representative of the main cloth remains highly disputed.

Evidence Quality Tier & Weight Category

Evidence Tier: A (Direct physical measurements)

Weight Category: High

Likelihood Estimates $P(E | H)$

$P(E | H_A) \approx 0.65 - 0.82$

Justification: A medieval origin is directly supported by the dating results. The clustering of dates from three independent laboratories strengthens H_A substantially, as this is the expected outcome if the cloth was manufactured in the 13th or 14th century.

$P(E | H_B) \approx 0.08 - 0.18$

Justification: An ancient origin would require the dated samples to be unrepresentative of the original cloth. While possible, this demands additional assumptions (such as invisible repair) that are not directly supported by the 1988 laboratory data itself.

$$P(E | H_C) \approx 0.08 - 0.18$$

Justification: Similar to H_B, an unknown physical mechanism would still require the cloth to be ancient. The 1988 results create significant tension unless the samples are shown to be non-representative.

$$P(E | H_D) \approx 0.08 - 0.18$$

Justification: A resurrection-linked origin would also require the tested samples to come from a later repair rather than the original 1st-century cloth. The dating results are difficult to reconcile without invoking sample contamination or reweaving.

Bayes Factors (midpoint estimates)

$$H_A \text{ vs } H_D \approx 5.8 \mid H_A \text{ vs } H_B \approx 5.5 \mid H_A \text{ vs } H_C \approx 5.5$$

Interpretation

Criterion 2.1 provides strong support for H_A. The consistent results from three independent laboratories represent one of the most direct challenges to any hypothesis requiring a 1st-century origin. While questions about sample location exist, the laboratory measurements themselves are robust. This criterion carries high weight (Tier A) and will exert significant downward pressure on H_B, H_C, and H_D unless later criteria in this domain successfully demonstrate that the samples were not representative.

Criterion 2.2: Sample Location and Selection Concerns

Evidence E

The 1988 radiocarbon samples were taken from a single corner area of the Shroud. Subsequent research has raised questions about whether this specific location is representative of the main body of the cloth. Studies by textile chemist Raymond Rogers and later researchers Benford and Marino noted differences in weave pattern, the presence of cotton fibers (not found in the main body), and chemical characteristics such as vanillin content in the sampled region compared to other areas, suggesting the possibility of medieval repair or reweaving in that section.

Strongest Alternative Counter

The sample location was carefully chosen by textile experts and Church authorities to avoid repaired or contaminated areas. Multiple lines of evidence (including microscopic

examination at the time) indicated that the sampled threads were original to the cloth, and no conclusive proof of reweaving in the exact sampled area has been universally accepted by the broader scientific community.

Contradiction Handling

While the official 1988 sampling protocol aimed to select original material, later chemical and microscopic studies have produced conflicting interpretations regarding the homogeneity of the corner area. These conflicting findings are treated as a legitimate source of uncertainty rather than dismissed. This criterion functions as an independent qualifier on how much evidential weight can be assigned to the 1988 radiocarbon measurements.

Expert Consensus Tracking

There is no clear consensus on this issue. Some researchers and textile specialists accept the samples as representative, while others (including several peer-reviewed papers) argue that the corner area likely contains medieval repairs. The question remains actively disputed.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Historical documentation + supporting chemical/microscopic studies)
Weight Category: High

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.35 - 0.50$$

Justification: If the cloth is medieval, concerns about reweaving in the corner are less critical, as even repaired material would still date to the medieval period. However, significant reweaving would still slightly complicate a straightforward medieval origin story.

$$P(E | H_B) \approx 0.25 - 0.45$$

Justification: An ancient origin becomes more plausible if the dated samples came from a later repair. The observed differences in fiber composition and chemistry provide moderate support for the possibility that the tested material is not original to a 1st-century cloth.

$$P(E | H_C) \approx 0.25 - 0.45$$

Justification: Similar to H_B. An unknown physical mechanism would also benefit if the radiocarbon samples do not represent the original cloth on which the image was formed.

$$P(E | H_D) \approx 0.25 - 0.45$$

Justification: A resurrection-linked origin would be significantly strengthened if the 1988 samples came from medieval repair work rather than the original cloth. The chemical and microscopic anomalies provide some support for this possibility.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 1.0 - 1.3 \mid H_B \text{ vs } H_A \approx 0.9 - 1.2 \mid H_C \text{ vs } H_A \approx 0.9 - 1.2$$

Interpretation

Criterion 2.2 introduces meaningful uncertainty into the radiocarbon results. While it does not overturn the 1988 laboratory measurements, it reduces their effective weight by raising legitimate questions about whether the dated material represents the original cloth. This criterion provides modest support for H_B, H_C, and H_D while slightly weakening H_A. It carries high weight (Tier C) because it directly affects how much evidential value can be assigned to Criterion 2.1.

Criterion 2.3: Reweaving Hypothesis

Evidence E

The main body of the Shroud of Turin is woven in a distinctive 3:1 herringbone twill pattern. This is a relatively complex weave that was known in the ancient world but is considered somewhat unusual for a 1st-century burial cloth in the region. The 1988 radiocarbon samples were taken from one corner of the cloth. Several researchers, including textile chemist Raymond Rogers and later researchers Benford and Marino, have argued that this corner section shows signs of medieval repair or reweaving. Supporting observations include the presence of cotton fibers (not found in the main body), differences in thread diameter, and possible stitching or weaving anomalies that suggest the area was rewoven at a later date.

Strongest Alternative Counter

The sampled corner area is original to the cloth and shows no conclusive evidence of reweaving. Differences in fiber content (such as occasional cotton) can be explained by contamination or minor variations during original manufacture. Textile experts involved in the 1988 sampling examined the area and determined it was suitable for dating, and no definitive proof of intentional medieval repair in the exact sampled threads has been universally accepted.

Contradiction Handling

While some microscopic and chemical studies have identified anomalies in the corner region, these findings have not been universally accepted. There is ongoing debate about whether the observed differences represent intentional medieval repair or natural variation/contamination. This lack of consensus is treated as a source of uncertainty. This criterion functions as an independent qualifier on the reliability of the 1988 radiocarbon results.

Expert Consensus Tracking

There is no strong consensus on the reweaving hypothesis. Some textile specialists and Shroud researchers support the possibility of repair in the sampled area, while others (including members of the original 1988 team) maintain that the samples were taken from original cloth. The issue remains disputed in the literature.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Historical documentation + microscopic/chemical studies)

Weight Category: High

Likelihood Estimates $P(E | H)$

$P(E | H_A) \approx 0.30 - 0.45$

Justification: If the cloth is medieval, localized repair or reweaving in one corner is historically plausible. However, it would still require explaining why only that specific area was repaired and why the repair material produced such consistent radiocarbon results.

$P(E | H_B) \approx 0.30 - 0.50$

Justification: An ancient origin becomes more plausible if the dated corner was rewoven in the medieval period. The reported differences in weave characteristics and

fiber content provide moderate support for the possibility that the tested material is not original.

$$P(E | H_C) \approx 0.30 - 0.50$$

Justification: Similar to H_B. An unknown physical mechanism would also benefit if the radiocarbon-dated section does not represent the original cloth on which the image was formed.

$$P(E | H_D) \approx 0.32 - 0.52$$

Justification: A resurrection-linked origin would be strengthened if the 1988 samples came from later repair work. The textile anomalies and reweaving claims provide some of the strongest support available for explaining the medieval radiocarbon date under this hypothesis.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 1.2 - 1.5 \mid H_B \text{ vs } H_A \approx 1.1 - 1.4 \mid H_C \text{ vs } H_A \approx 1.1 - 1.4$$

Interpretation

Criterion 2.3 introduces further uncertainty regarding the reliability of the 1988 dating results. While the reweaving hypothesis is not proven, the documented differences in the corner area (weave characteristics and fiber content) provide reasonable grounds to question whether the dated samples represent the original cloth. This criterion modestly weakens H_A and provides mild support for H_B, H_C, and H_D. It carries high weight (Tier C) because it directly impacts how much evidential value can be assigned to the radiocarbon measurements.

Criterion 2.4: Statistical Heterogeneity and Contamination Hypotheses

Evidence E

Some researchers have raised concerns about statistical heterogeneity in the 1988 radiocarbon data. Analyses of the published results (including work by statisticians examining the inter-laboratory variation) have suggested that the variation between the subsamples from different laboratories was greater than would normally be expected from purely random measurement error. Additionally, contamination hypotheses

propose that newer carbon (from handling, candle smoke, bacteria, or other sources) could have been absorbed into the linen fibers over centuries, potentially making an older cloth appear younger than it actually is.

Strongest Alternative Counter

The 1988 results showed good overall consistency between the three laboratories. Standard pretreatment procedures were used to remove potential contaminants, and the final combined date range was considered reliable by the participating labs and the journal *Nature*. Claims of significant statistical heterogeneity or meaningful contamination have not been widely accepted by the broader scientific community as overturning the primary results.

Contradiction Handling

While the laboratories applied standard cleaning protocols, it is difficult to completely rule out all forms of contamination over the long history of the cloth. Similarly, debates about statistical heterogeneity in the data remain unresolved in the broader literature. These issues are treated as sources of uncertainty rather than definitive refutations of the dating.

Expert Consensus Tracking

The core 1988 dating results are generally accepted as technically competent. However, claims of statistical anomalies and contamination effects remain disputed, with some independent statisticians and Shroud researchers arguing for greater caution in interpreting the precision of the final date range.

Evidence Quality Tier & Weight Category

Evidence Tier: A (Direct physical measurements) + some statistical re-analysis (Tier C)
Weight Category: Medium-High

Likelihood Estimates $P(E | H)$

$P(E | H_A) \approx 0.45 - 0.60$

Justification: A medieval origin aligns well with the dating results. While some statistical variation exists, it does not strongly contradict a 13th–14th century date.

Contamination would more likely make the cloth appear older rather than younger in most scenarios.

$$P(E | H_B) \approx 0.20 - 0.35$$

Justification: An ancient origin would require either significant contamination with younger carbon or that the statistical variation indicates problems with the sample or measurement process. Both possibilities introduce additional assumptions.

$$P(E | H_C) \approx 0.20 - 0.35$$

Justification: Similar to H_B. An unknown physical mechanism would also require the radiocarbon results to be misleading due to contamination or sampling issues.

$$P(E | H_D) \approx 0.20 - 0.38$$

Justification: A resurrection-linked origin would benefit if contamination or statistical issues undermined the reliability of the medieval date. However, the degree of contamination needed to shift a 1st-century date into the 13th–14th century range is considered substantial by most experts.

Bayes Factors (midpoint estimates)

$$H_A \text{ vs } H_D \approx 2.2 - 2.8 \mid H_A \text{ vs } H_B \approx 2.0 - 2.5 \mid H_A \text{ vs } H_C \approx 2.0 - 2.5$$

Interpretation

Criterion 2.4 provides moderate support for H_A. While concerns about statistical heterogeneity and potential contamination exist, they have not been strong enough to overturn the primary dating results in the eyes of most researchers. This criterion reduces (but does not eliminate) the impact of the radiocarbon evidence against H_B, H_C, and H_D. It carries Medium-High weight because it qualifies, rather than fully supports or refutes, the strength of the 1988 dating. This criterion functions as an independent qualifier on how much evidential weight can be confidently assigned to Criterion 2.1.

Criterion 2.5: Lack of Modern Independent Retesting

Evidence E

No new, independent radiocarbon dating of the Shroud of Turin has been conducted since the 1988 tests. Despite repeated proposals from scientists and requests for access, the custodians of the Shroud have not permitted additional radiocarbon testing

using modern techniques. Current AMS (Accelerator Mass Spectrometry) methods require significantly smaller samples and offer improved pretreatment protocols compared to those available in 1988.

Strongest Alternative Counter

The 1988 dating was rigorously performed by three independent laboratories using the best methods available at the time. Additional testing is unnecessary because the original results were consistent and the cloth has already been dated with sufficient precision. Further testing could risk damaging the relic without providing meaningful new information, and many experts consider the existing data adequate for historical purposes.

Contradiction Handling

The refusal to allow new testing creates a significant limitation in the current evidence base. While the 1988 results stand, the inability to verify them with modern methods (smaller samples, better contamination removal, and improved calibration) leaves open the possibility that new data could alter or refine the dating conclusions. This limitation is treated as a genuine weakness in the overall radiocarbon evidence.

Expert Consensus Tracking

There is broad agreement among scientists that retesting with modern methods would be scientifically valuable. However, opinions differ on whether the lack of retesting significantly undermines the 1988 results or simply reflects careful stewardship of the artifact.

Evidence Quality Tier & Weight Category

Evidence Tier: E (Interpretive inference regarding the state of evidence)

Weight Category: Medium

Likelihood Estimates $P(E | H)$

$P(E | H_A) \approx 0.40 - 0.55$

Justification: A medieval origin is already supported by the existing dating. The lack of new testing does not strongly harm H_A , though it prevents the possibility of further confirmation using improved methods.

$$P(E | H_B) \approx 0.25 - 0.40$$

Justification: An ancient origin would benefit from the opportunity for retesting with modern techniques that could better detect or remove contamination. The current inability to perform such tests leaves open a pathway that could potentially support an older date.

$$P(E | H_C) \approx 0.25 - 0.40$$

Justification: Similar to H_B. Modern retesting could potentially strengthen or weaken the case for an unknown physical mechanism depending on the outcome.

$$P(E | H_D) \approx 0.28 - 0.45$$

Justification: A resurrection-linked origin would gain from the chance to retest with contemporary methods, as improved protocols might resolve or reduce concerns about the medieval date. The lack of retesting keeps this possibility open.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 0.8 - 1.1 \mid H_B \text{ vs } H_A \approx 0.7 - 1.0 \mid H_C \text{ vs } H_A \approx 0.7 - 1.0$$

Interpretation

Criterion 2.5 represents a limitation rather than positive evidence. The absence of modern independent retesting slightly weakens the overall confidence that can be placed in the 1988 results. While it does not actively support any hypothesis, it prevents H_A from gaining further confirmation and leaves more room for H_B, H_C, and H_D than would exist if retesting had occurred and upheld the original date. This criterion carries Medium weight (Tier E) because it qualifies the strength of the radiocarbon evidence rather than directly supporting or refuting any specific formation hypothesis. It functions as an independent qualifier on how much evidential weight can be confidently assigned to the 1988 results overall.

Domain 2 Summary: Radiocarbon Dating & Sampling Integrity

Domain Overview

Domain 2 examines the 1988 radiocarbon dating results and the surrounding questions about sample integrity. This is the strongest skeptical domain in this evaluation and directly challenges hypotheses requiring a 1st-century origin.

Criteria Summary

Criterion	Description	Weight	Evidence Tier	Primary Impact on Hypotheses
2.1	1988 radiocarbon results & consistency	High	A	Strongly supports H_A
2.2	Sample location and selection concerns	High	C	Moderately weakens H_A
2.3	Reweaving hypothesis	High	C	Moderately weakens H_A
2.4	Statistical heterogeneity & contamination	Medium-High	A	Moderately supports H_A
2.5	Lack of modern independent retesting	Medium	E	Slightly weakens H_A

Hypothesis Performance in Domain 2

H_A (Medieval Artifact): Performs strongest in this domain. Criterion 2.1 provides strong support, though Criteria 2.2 and 2.3 introduce meaningful uncertainty regarding sample representativeness.

H_B, H_C, and H_D: All face significant pressure from the 1988 dating results. However, questions about sample location and reweaving (2.2 & 2.3) provide moderate relief by raising the possibility that the dated material may not represent the original cloth.

Overall Domain Impact: Strong net support for H_A, but with notable qualifications due to sampling concerns.

Key Findings

- The 1988 dating remains the single strongest piece of evidence favoring a medieval origin.
- Concerns about the sample location and possible reweaving meaningfully reduce

the weight that can be confidently assigned to the radiocarbon results.

- The lack of modern retesting leaves some room for future revision of the dating evidence.

Implications

Domain 2 creates the strongest counter-pressure against H_B, H_C, and H_D seen so far. However, it does not fully neutralize them due to unresolved questions about sample integrity. This domain will be one of the most important factors in the final comparative assessment.

Dependency Considerations in Domain 2

Criteria 2.2 (Sample Location), 2.3 (Reweaving Hypothesis), and 2.4 (Statistical Heterogeneity & Contamination) are not fully independent. Concerns about reweaving directly influence the weight that can be assigned to sample location and heterogeneity claims. The Monte Carlo model in Domain 7 incorporates domain-level correlations to partially account for this interdependence. Sensitivity testing (see Section 7.3) was also used to evaluate how much the overall results shift when the effective weight of these interrelated criteria is varied. This approach reduces but does not eliminate the risk of over-weighting correlated lines of evidence against the 1988 radiocarbon results.

Domain 2 Criterion Registry

This table presents the complete, auditable input layer for Domain 2. These values are used directly by the weighted aggregation function to produce the domain-level likelihood ranges.

ID	Criterion	Tier	Weight	H_A Range	H_B Range	H_C Range	H_D Range
2.1	1988 radiocarbon results and inter-laboratory consistency	A	High	0.65 – 0.82	0.08 – 0.18	0.08 – 0.18	0.08 – 0.18
2.2	Sample location and selection	C	High	0.35 –	0.25 –	0.25 –	0.25 –

	concerns			0.50	0.45	0.45	0.45
2.3	Reweaving hypothesis	C	High	0.30 – 0.45	0.30 – 0.50	0.30 – 0.50	0.32 – 0.52
2.4	Statistical heterogeneity and contamination hypotheses	A	Medium-High	0.45 – 0.60	0.20 – 0.35	0.20 – 0.35	0.20 – 0.38
2.5	Lack of modern independent retesting	E	Medium	0.40 – 0.55	0.25 – 0.40	0.25 – 0.40	0.28 – 0.45

Derived Domain 2 Likelihood Ranges (Weighted Aggregation)

Hypothesis	Derived Range	Notes
H_A (Medieval Artifact)	0.454 – 0.610	Strong support from Criterion 2.1, moderated by 2.2–2.5
H_B (Natural Ancient)	0.199 – 0.355	Significant pressure from 2.1, partial relief from 2.2–2.3
H_C (Unknown Physical)	0.199 – 0.355	Significant pressure from 2.1, partial relief from 2.2–2.3
H_D (Resurrection-linked)	0.205 – 0.370	Significant pressure from 2.1, partial relief from 2.2–2.3

These derived ranges feed directly into the Monte Carlo synthesis in Domain 7.

Domain 3: Forensic / Medical Coherence

Domain Overview

Domain 3 evaluates the anatomical, wound, and post-mortem features visible on the Shroud. It assesses how well the image reflects realistic crucifixion trauma and medical knowledge.

Criterion 3.1: Anatomical Accuracy of Wounds and Body Positioning

Evidence E

The wounds and body positioning visible on the Shroud image demonstrate a high degree of anatomical and medical accuracy. The wounds align with known Roman crucifixion practices, including the placement of nail wounds through the wrists (rather than palms), scourge marks consistent with Roman flagra, and overall body posture that matches the mechanics of crucifixion. The image shows realistic wound patterns, including blood flow directions that correspond to the effects of gravity on a crucified body.

Strongest Alternative Counter

A skilled medieval artist or forger with access to knowledge of anatomy and crucifixion could have created a reasonably accurate representation. Artistic conventions and available medical knowledge in the 13th–14th century, combined with observation of actual crucifixions or descriptions in texts, could account for the observed accuracy without requiring modern forensic understanding. However, the level of consistent detail across subtle features (wound angles, blood flow directions, and body mechanics) would still require an unusually high degree of anatomical knowledge and observational skill for the period.

Contradiction Handling

While medieval artists had some knowledge of crucifixion, the level of detailed anatomical and wound consistency across the entire body (including subtle details such as wound angles and blood flow) exceeds what would typically be expected from artistic convention alone. This tension is noted but does not fully eliminate the possibility of skilled artistic recreation. This criterion functions as an independent forensic constraint on possible image-formation mechanisms.

Expert Consensus Tracking

Forensic pathologists and medical examiners who have studied the Shroud generally acknowledge the high degree of anatomical accuracy in the wound patterns and body

positioning. There is broad agreement that the image reflects realistic crucifixion trauma, though interpretations of how the image was formed vary.

Evidence Quality Tier & Weight Category

Evidence Tier: D (Expert testimony / consensus)

Weight Category: High

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.20 - 0.35$$

Justification: A medieval artist would need substantial anatomical knowledge and careful observation to accurately depict the full range of crucifixion wounds and body mechanics. While possible for a highly skilled individual, the consistency and detail across the entire image make this more difficult than simpler artistic representations.

$$P(E | H_B) \approx 0.45 - 0.65$$

Justification: An actual crucifixion victim would naturally produce anatomically accurate wound patterns and body positioning consistent with the mechanics of Roman crucifixion. This hypothesis aligns well with the observed forensic details.

$$P(E | H_C) \approx 0.45 - 0.65$$

Justification: Similar to H_B . An unknown physical mechanism forming the image on an actual crucified body would also be expected to accurately record the real anatomical and wound features.

$$P(E | H_D) \approx 0.50 - 0.70$$

Justification: A resurrection-linked process acting on an actual crucified body would be expected to preserve or record the accurate anatomical and wound details. This hypothesis is highly compatible with the forensic accuracy observed.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 2.8 - 3.5 \mid H_B \text{ vs } H_A \approx 2.3 - 3.0 \mid H_C \text{ vs } H_A \approx 2.3 - 3.0$$

Interpretation

Criterion 3.1 provides meaningful support for hypotheses involving an actual crucified body (H_B , H_C , and H_D). The high degree of anatomical and forensic accuracy

significantly weakens H_A, as it would require an unusually skilled and knowledgeable medieval artist. This criterion carries high weight (Tier D) and strengthens the case that the image reflects real crucifixion trauma rather than purely artistic invention.

Criterion 3.2: Blood Flow Consistency and Directionality

Evidence E

The bloodstains on the Shroud show patterns consistent with realistic blood flow under the influence of gravity. There are distinct flows from the wrist wounds, feet, and side wound. Some blood flows appear to have occurred while the body was in a vertical position (during crucifixion), while others are consistent with a horizontal position (after the body was removed from the cross and laid flat). The direction and pooling of blood align with known physiological behavior.

Strongest Alternative Counter

A skilled medieval artist could have painted or applied blood-like substances in patterns that appear realistic. Artistic conventions or direct observation of wounds and blood flow could have been used to create convincing directional patterns without requiring actual physiological events. However, achieving consistent realism across multiple directional flows that correspond to both vertical (crucifixion) and horizontal (post-crucifixion) body positions simultaneously would still require a high level of anatomical and physiological knowledge.

Contradiction Handling

While artistic skill can produce realistic-looking blood flows, the combination of multiple directional flows that correspond to both vertical and horizontal body positions across the entire image adds complexity that would be difficult to consistently achieve through artistic means alone. This tension is acknowledged. This criterion functions as an independent forensic constraint on possible image-formation mechanisms.

Expert Consensus Tracking

Forensic and medical analysts who have examined the blood flow patterns generally consider them to be consistent with real crucifixion and post-crucifixion handling. There is reasonable agreement on the physiological plausibility of the flows, though debate remains about how the image and bloodstains were formed.

Evidence Quality Tier & Weight Category

Evidence Tier: D (Expert testimony / consensus)

Weight Category: High

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.15 - 0.28$$

Justification: Creating accurate, multi-directional blood flows that correspond to both vertical and horizontal positions would require significant anatomical and physiological knowledge. While a highly skilled artist could approximate this, achieving consistent realism across the full image is considered difficult.

$$P(E | H_B) \approx 0.48 - 0.68$$

Justification: An actual crucified body would naturally produce blood flows that reflect changes in body position (vertical during crucifixion, then horizontal afterward). This hypothesis aligns well with the observed patterns.

$$P(E | H_C) \approx 0.48 - 0.68$$

Justification: Similar to H_B . An unknown physical mechanism forming the image on a real body would also be expected to accurately record realistic blood flow patterns.

$$P(E | H_D) \approx 0.52 - 0.72$$

Justification: A resurrection-linked process acting on an actual body would preserve or record the authentic blood flow patterns resulting from crucifixion and subsequent handling. This is highly compatible with the evidence.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 3.5 - 4.2 \mid H_B \text{ vs } H_A \approx 3.0 - 3.7 \mid H_C \text{ vs } H_A \approx 3.0 - 3.7$$

Interpretation

Criterion 3.2 adds further support for hypotheses involving a real crucified body. The realistic, position-dependent blood flow patterns significantly weaken H_A while strengthening H_B , H_C , and H_D . This criterion carries high weight (Tier D) and contributes to the overall forensic coherence of the image as reflecting actual trauma rather than artistic creation.

Criterion 3.3: Roman Scourging Pattern Match

Evidence E

The Shroud image displays numerous small, dumbbell-shaped wounds across the back, shoulders, and legs, consistent with scourging by a Roman flagrum (a whip with two or three leather thongs ending in small weights or bones). The pattern and distribution of the wounds suggest they were inflicted by two individuals standing on either side of the victim, with the strikes following a consistent anatomical pattern.

Strongest Alternative Counter

A medieval artist with knowledge of Roman history or access to descriptions of crucifixion could have replicated scourge marks based on textual accounts or artistic tradition. The dumbbell shape could have been achieved through careful painting or stamping techniques. However, reproducing the precise anatomical distribution, consistent dumbbell shape, and bilateral pattern across the full body would still require a high level of detailed historical and anatomical knowledge plus significant artistic skill.

Contradiction Handling

While knowledge of Roman scourging existed in the medieval period, the precise anatomical distribution, consistent dumbbell shape, and bilateral pattern across the body would require a high level of detailed knowledge and artistic skill to reproduce accurately without direct reference to an actual victim. This adds some tension to purely artistic explanations. This criterion functions as an independent forensic constraint on possible image-formation mechanisms.

Expert Consensus Tracking

Forensic pathologists and historians who have studied the scourge marks generally agree that they are consistent with Roman flagellation practices. The dumbbell-shaped wounds and bilateral pattern are considered medically and historically plausible.

Evidence Quality Tier & Weight Category

Evidence Tier: D (Expert testimony / consensus)

Weight Category: Medium-High

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.18 - 0.32$$

Justification: Reproducing the specific dumbbell-shaped wounds and anatomically consistent bilateral pattern would require detailed knowledge of Roman scourging instruments and practices. While possible for a skilled and informed artist, it adds complexity to a purely medieval artistic explanation.

$$P(E | H_B) \approx 0.50 - 0.70$$

Justification: An actual victim of Roman scourging would naturally exhibit wounds matching the historical instrument and typical patterns of infliction. This hypothesis aligns well with the observed evidence.

$$P(E | H_C) \approx 0.50 - 0.70$$

Justification: Similar to H_B. An unknown physical mechanism forming the image on a real scourged body would accurately record the wound patterns.

$$P(E | H_D) \approx 0.55 - 0.75$$

Justification: A resurrection-linked process acting on an actual body would preserve the authentic scourge wounds inflicted during Roman punishment. This is highly compatible with the evidence.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 3.0 - 3.8 \mid H_B \text{ vs } H_A \approx 2.6 - 3.3 \mid H_C \text{ vs } H_A \approx 2.6 - 3.3$$

Interpretation

Criterion 3.3 provides additional support for hypotheses involving a real crucified and scourged body. The specific characteristics of the scourge marks (dumbbell shape and bilateral pattern) moderately weaken H_A while strengthening H_B, H_C, and H_D. This criterion carries Medium-High weight (Tier D) and contributes to the cumulative forensic case for authenticity of the trauma patterns.

Criterion 3.4: Wrist vs. Palm Nailing Placement

Evidence E

The nail wounds visible on the Shroud are located in the wrists rather than the palms. Specifically, the wounds align with the area known as Destot's space in the wrist, which

is anatomically capable of supporting the full weight of a crucified body. In contrast, nails driven through the palms would typically tear out under body weight. This detail contradicts traditional artistic depictions, which often show nails through the palms.

Strongest Alternative Counter

A medieval artist with access to biblical descriptions, early Church traditions, or limited anatomical knowledge could have placed the wounds in the wrists by chance or based on some available references. Artistic variation existed, and not all medieval depictions strictly followed palm nailing. However, the dominant tradition in medieval Christian art showed palm nailing, making accurate wrist placement a notable departure from convention.

Contradiction Handling

While some artistic flexibility existed, the dominant tradition in medieval Christian art showed palm nailing. Placing the wounds accurately in the wrists (a detail not widely understood until modern medical analysis) would require either unusual knowledge or direct reference to an actual crucified body. This creates meaningful tension with a purely artistic explanation. This criterion functions as an independent forensic constraint on possible image-formation mechanisms.

Expert Consensus Tracking

Forensic pathologists and medical researchers who have studied crucifixion generally agree that wrist nailing (specifically through Destot's space) is the anatomically correct method for supporting body weight. The Shroud's wound placement is considered consistent with this understanding.

Evidence Quality Tier & Weight Category

Evidence Tier: D (Expert testimony / consensus)

Weight Category: High

Likelihood Estimates $P(E | H)$

$P(E | H_A) \approx 0.10 - 0.22$

Justification: Most medieval artistic representations depicted nails through the palms. Creating accurate wrist nailing that aligns with modern forensic understanding would require either exceptional anatomical insight or direct observation of real crucifixion wounds, both of which are considered unlikely for a typical medieval artist.

$$P(E | H_B) \approx 0.55 - 0.75$$

Justification: An actual crucifixion would produce nail wounds in the wrists, as this is the mechanically sound location. This hypothesis aligns strongly with the anatomical evidence.

$$P(E | H_C) \approx 0.55 - 0.75$$

Justification: Similar to H_B . An unknown physical mechanism forming the image on a real body would accurately record the correct anatomical wound location.

$$P(E | H_D) \approx 0.60 - 0.80$$

Justification: A resurrection-linked process acting on an actual crucified body would preserve the authentic wrist nailing wounds. This is highly compatible with the forensic evidence.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 4.5 - 5.5 \mid H_B \text{ vs } H_A \approx 3.8 - 4.8 \mid H_C \text{ vs } H_A \approx 3.8 - 4.8$$

Interpretation

Criterion 3.4 provides strong support for hypotheses involving a real crucifixion. The anatomically correct wrist nailing significantly weakens H_A , as it contradicts common medieval artistic convention. This criterion carries high weight (Tier D) and represents one of the clearer forensic indicators favoring an actual historical event over artistic creation.

Criterion 3.5: Side Wound Characteristics

Evidence E

The Shroud shows a large wound on the right side of the chest, consistent with a spear thrust. The wound is located between the fifth and sixth ribs and appears to have penetrated the chest cavity. There is evidence of blood flow that includes both whole blood and a clearer fluid, which some analyses interpret as serum separating from clotted blood — a process that occurs after death. The angle and location of the wound are consistent with Roman spear usage during crucifixion.

Strongest Alternative Counter

A medieval artist could have depicted a side wound based on biblical descriptions (John 19:34) and artistic tradition. The addition of serum-like fluid could be achieved through artistic technique or the use of different pigments or liquids. However, accurately reproducing the specific post-mortem serum separation and precise anatomical location would still require more advanced medical understanding than was typical for medieval artists.

Contradiction Handling

While the biblical account mentions a spear wound, the specific medical details — such as the separation of serum from blood and the precise anatomical location — go beyond simple artistic representation. Reproducing these subtle post-mortem characteristics convincingly would require either advanced knowledge or direct reference to actual trauma. This criterion functions as an independent forensic constraint on possible image-formation mechanisms.

Expert Consensus Tracking

Forensic pathologists who have examined the side wound generally consider its location, size, and associated fluid patterns to be consistent with a post-mortem spear thrust into the chest cavity. The presence of both blood and serum is viewed as medically plausible for a wound inflicted after death.

Evidence Quality Tier & Weight Category

Evidence Tier: D (Expert testimony / consensus)

Weight Category: Medium-High

Likelihood Estimates $P(E | H)$

$P(E | H_A) \approx 0.15 - 0.28$

Justification: While medieval artists knew the biblical account of the spear wound, accurately depicting the post-mortem separation of serum and the specific anatomical details would require more advanced medical understanding than was typical. This adds difficulty to a purely artistic explanation.

$P(E | H_B) \approx 0.48 - 0.68$

Justification: An actual post-mortem spear wound on a crucified body would naturally produce blood and serum separation. This hypothesis aligns well with the observed

characteristics.

$$P(E | H_C) \approx 0.48 - 0.68$$

Justification: Similar to H_B. An unknown physical mechanism forming the image on a real body would accurately record the wound and associated fluid patterns.

$$P(E | H_D) \approx 0.52 - 0.72$$

Justification: A resurrection-linked process acting on an actual body would preserve the authentic side wound and post-mortem fluid characteristics. This is highly compatible with the evidence.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 3.2 - 4.0 \mid H_B \text{ vs } H_A \approx 2.8 - 3.5 \mid H_C \text{ vs } H_A \approx 2.8 - 3.5$$

Interpretation

Criterion 3.5 provides additional forensic support for hypotheses involving a real body. The characteristics of the side wound, including the apparent serum separation, moderately weaken H_A while strengthening H_B, H_C, and H_D. This criterion carries Medium-High weight (Tier D) and contributes to the cumulative medical coherence of the trauma patterns on the Shroud.

Criterion 3.6: Post-Mortem Indicators

Evidence E

The Shroud image shows several features consistent with post-mortem changes. These include:

- Body positioning and limb rigidity suggestive of rigor mortis.
- Blood flows (particularly from the side wound) that appear to have occurred after death, when the body was in a horizontal position.
- Lack of signs of significant decomposition despite the time between death and burial.

These features align with known physiological processes that occur after death.

Strongest Alternative Counter

A skilled medieval artist could have depicted a body in rigor mortis or with post-mortem blood flows based on observation of deceased individuals or artistic convention. These details could be added intentionally to increase realism. However, consistently applying multiple post-mortem indicators across the entire image in a medically accurate manner (especially the interaction between wound position and body posture) would still require notable skill and attention to detail.

Contradiction Handling

While artists could observe and depict rigor mortis and post-mortem blood flow, consistently applying these details across the entire image in a medically accurate manner (especially the interaction between wound position and body posture) adds complexity to an artistic explanation. This criterion functions as an independent forensic constraint on possible image-formation mechanisms.

Expert Consensus Tracking

Forensic pathologists generally agree that several features on the Shroud (including body rigidity and certain blood flow patterns) are consistent with post-mortem changes. These observations are considered medically plausible.

Evidence Quality Tier & Weight Category

Evidence Tier: D (Expert testimony / consensus)

Weight Category: Medium

Likelihood Estimates $P(E | H)$

$P(E | H_A) \approx 0.22 - 0.38$

Justification: A medieval artist could incorporate post-mortem features based on observation. However, achieving consistent medical accuracy across multiple indicators simultaneously would require notable skill and attention to detail.

$P(E | H_B) \approx 0.48 - 0.68$

Justification: An actual deceased body would naturally exhibit rigor mortis and post-mortem blood flows. This hypothesis aligns well with the observed indicators.

$P(E \mid H_C) \approx 0.48 - 0.68$

Justification: Similar to H_B. An unknown physical mechanism forming the image on a real body would accurately record post-mortem changes.

$P(E \mid H_D) \approx 0.52 - 0.72$

Justification: A resurrection-linked process acting on an actual body would preserve authentic post-mortem features. This is highly compatible with the evidence.

Bayes Factors (midpoint estimates)

$H_D \text{ vs } H_A \approx 2.4 - 3.0 \mid H_B \text{ vs } H_A \approx 2.1 - 2.7 \mid H_C \text{ vs } H_A \approx 2.1 - 2.7$

Interpretation

Criterion 3.6 provides moderate support for hypotheses involving a real body. The presence of post-mortem indicators (rigor mortis and post-death blood flows) weakens H_A to a moderate degree while strengthening H_B, H_C, and H_D. This criterion carries Medium weight (Tier D) and adds to the overall forensic coherence of the image as reflecting a real deceased individual.

Domain 3 Summary: Forensic / Medical Coherence

Domain Overview

Domain 3 evaluates the anatomical, wound, and post-mortem features visible on the Shroud. It assesses how well the image reflects realistic crucifixion trauma and medical knowledge.

Criteria Summary

Criterion	Description	Weight	Evidence Tier	Primary Impact on Hypotheses
3.1	Anatomical accuracy of wounds and positioning	High	D	Strongly weakens H_A
3.2	Blood flow consistency and directionality	High	D	Strongly weakens H_A
	Roman scourging pattern	Medium-		Moderately

3.3	match	High	D	weakens H_A
3.4	Wrist vs. palm nailing placement	High	D	Strongly weakens H_A
3.5	Side wound characteristics	Medium-High	D	Moderately weakens H_A
3.6	Post-mortem indicators	Medium	D	Moderately weakens H_A

Hypothesis Performance in Domain 3

H_A (Medieval Artifact): Performs weakest in this domain. Multiple high-weight criteria (especially 3.1, 3.2, and 3.4) significantly challenge the idea that a medieval artist could have accurately reproduced the full range of forensic details.

H_B, H_C, and H_D: All perform well. The forensic and medical accuracy of the wounds and body features is highly consistent with an actual crucified body.

Overall Domain Impact: Strong net support for H_B, H_C, and H_D, with H_D showing a slight edge due to the cumulative precision of the details.

Key Findings

The Shroud displays a high level of forensic and anatomical accuracy that is difficult to explain through conventional medieval artistry. Several specific details (wrist nailing, blood flow patterns, and scourge marks) align closely with known Roman crucifixion practices. This domain provides some of the strongest cumulative evidence against a purely artistic origin.

Implications

Domain 3 significantly strengthens the case that the image reflects real trauma from crucifixion. It creates strong counter-pressure against H_A and supports the hypotheses that involve an actual body (H_B, H_C, and H_D). This domain will play an important role in the final synthesis.

Domain 3 Criterion Registry

This table presents the complete, auditable input layer for Domain 3. These values are used directly by the weighted aggregation function to produce the domain-level likelihood ranges.

ID	Criterion	Tier	Weight	H_A Range	H_B Range	H_C Range	H_D Range
3.1	Anatomical accuracy of wounds and positioning	D	High	0.20 – 0.35	0.45 – 0.65	0.45 – 0.65	0.50 – 0.70
3.2	Blood flow consistency and directionality	D	High	0.15 – 0.28	0.48 – 0.68	0.48 – 0.68	0.52 – 0.72
3.3	Roman scourging pattern match	D	Medium-High	0.18 – 0.32	0.50 – 0.70	0.50 – 0.70	0.55 – 0.75
3.4	Wrist vs. palm nailing placement	D	High	0.10 – 0.22	0.55 – 0.75	0.55 – 0.75	0.60 – 0.80
3.5	Side wound characteristics	D	Medium-High	0.15 – 0.28	0.48 – 0.68	0.48 – 0.68	0.52 – 0.72
3.6	Post-mortem indicators	D	Medium	0.22 – 0.38	0.48 – 0.68	0.48 – 0.68	0.52 – 0.72

Derived Domain 3 Likelihood Ranges (Weighted Aggregation)

Hypothesis	Derived Range	Notes
H_A (Medieval Artifact)	0.110 – 0.271	Strongly constrained by multiple high-weight forensic criteria
H_B (Natural Ancient)	0.460 – 0.699	Strong compatibility with real crucifixion trauma

H_C (Unknown Physical)	0.460 – 0.699	Strong compatibility with real crucifixion trauma
H_D (Resurrection-linked)	0.539 – 0.787	Highest compatibility with cumulative forensic precision

These derived ranges feed directly into the Monte Carlo synthesis in Domain 7.

Domain 4: Comparative Artifact Analysis

Domain Overview

Domain 4 evaluates how the Shroud compares to other known medieval relics, artistic techniques, and modern reproduction attempts. It assesses whether the Shroud is an outlier or fits within expected categories of historical artifacts.

Criterion 4.1: Comparison with Known Medieval Relics and Image Techniques

Evidence E

The Shroud of Turin differs significantly from other known medieval relics and image-bearing artifacts. Most medieval religious images were created using pigments, binders, or contact printing methods that leave visible residues or directional marks. In contrast, the Shroud shows no pigments, no binders, extreme superficiality, and accurate 3D encoding — a combination not found in comparable medieval relics or artworks.

Strongest Alternative Counter

The Shroud could still be a medieval creation that used an unusual or experimental technique not commonly preserved in other artifacts. Some experimental methods (such as contact prints or low-energy chemical processes) have produced partial similarities, suggesting that a skilled medieval artisan might have used a rare or now-lost method. However, no documented medieval technique has successfully replicated the full combination of extreme superficiality, accurate non-contact 3D encoding, and complete absence of pigments or directionality.

Contradiction Handling

While experimental reproductions have achieved limited success with individual features, no medieval or modern technique has successfully replicated the full

combination of properties (superficiality, lack of pigments, 3D encoding, and absence of scorching) in a single coherent image. This gap is acknowledged as a limitation for purely artistic explanations. This criterion functions as an independent comparative constraint on possible image-formation mechanisms.

Expert Consensus Tracking

Researchers generally agree that the Shroud is unique among known medieval relics in its combination of physical characteristics. While some experimental work has produced partial matches, the full set of features remains unmatched by documented medieval techniques.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Historical documentation + comparative analysis)

Weight Category: Medium-High

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.15 - 0.28$$

Justification: The Shroud's combination of features is not consistent with typical medieval relic production or known artistic techniques. While an unusual method cannot be completely ruled out, the lack of comparable examples from the period reduces the likelihood under H_A .

$$P(E | H_B) \approx 0.35 - 0.55$$

Justification: An ancient burial cloth could plausibly differ from typical medieval relics. However, natural formation mechanisms would still need to explain why this particular cloth developed such unusual properties compared to other ancient textiles.

$$P(E | H_C) \approx 0.45 - 0.65$$

Justification: An unknown physical mechanism could produce image characteristics that differ from conventional medieval or artistic methods. This hypothesis is reasonably compatible with the Shroud's uniqueness.

$$P(E | H_D) \approx 0.48 - 0.68$$

Justification: A resurrection-linked process could result in image properties that do not match conventional relic or artistic production. This is consistent with the observed differences from other medieval artifacts.

Bayes Factors (midpoint estimates)

H_D vs $H_A \approx 3.0 - 3.8$ | H_C vs $H_A \approx 2.7 - 3.4$ | H_B vs $H_A \approx 2.0 - 2.7$

Interpretation

Criterion 4.1 shows that the Shroud is an outlier compared to known medieval relics and image techniques. This moderately weakens H_A while providing moderate support for H_C and H_D . This criterion carries Medium-High weight (Tier C) and contributes to the case that the Shroud does not fit easily within conventional medieval artifact categories.

Criterion 4.2: Statistical Uniqueness of the Shroud Image Properties

Evidence E

When the full set of physical characteristics of the Shroud image is considered together (extreme superficiality, absence of pigments and directionality, accurate 3D encoding, lack of scorching, and chemical alteration of the fibers), the combination is highly unusual. No other known historical image or relic displays this specific cluster of properties. Statistical analyses of image characteristics have suggested that the Shroud is an outlier when compared to both medieval artworks and other ancient textiles.

Strongest Alternative Counter

While the combination of features is unusual, it is not impossible for a medieval creator to have produced through an experimental or serendipitous process. Rarity alone does not prove non-medieval origin, as unique artifacts can still be created by human hands under specific conditions. However, the specific cluster of properties (extreme superficiality + accurate non-contact 3D encoding + complete absence of pigments/directionality) has no clear parallel in surviving medieval artifacts or experimental reproductions.

Contradiction Handling

The statistical uniqueness of the combined features increases the difficulty of explaining the Shroud through conventional medieval techniques. However, uniqueness by itself does not rule out an unknown or rare medieval method. This limitation is acknowledged. This criterion functions as an independent comparative constraint on possible image-formation mechanisms.

Expert Consensus Tracking

Most researchers agree that the Shroud possesses a rare combination of image properties not commonly found in other artifacts. There is less agreement on how much weight this uniqueness should carry in determining its origin.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Comparative analysis + statistical considerations)

Weight Category: Medium

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.12 - 0.25$$

Justification: The combination of features is statistically unusual for medieval image production. While not impossible, it would require a rare or experimental technique that has no clear parallel in surviving artifacts.

$$P(E | H_B) \approx 0.30 - 0.50$$

Justification: An ancient cloth could develop unusual properties through natural processes, though the specific combination observed remains uncommon even among ancient textiles.

$$P(E | H_C) \approx 0.45 - 0.65$$

Justification: An unknown physical mechanism could produce a statistically unusual set of image characteristics. This hypothesis is reasonably compatible with the observed uniqueness.

$$P(E | H_D) \approx 0.48 - 0.68$$

Justification: A resurrection-linked process could result in image properties that differ significantly from conventional formation methods, making the statistical uniqueness more expected under this hypothesis.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 3.2 - 4.0 \mid H_C \text{ vs } H_A \approx 2.8 - 3.5 \mid H_B \text{ vs } H_A \approx 2.0 - 2.6$$

Interpretation

Criterion 4.2 shows that the Shroud's image properties are statistically unusual when compared to other known artifacts. This moderately weakens H_A and provides moderate support for H_C and H_D. This criterion carries Medium weight (Tier C) because uniqueness, while notable, must be interpreted alongside other evidence rather than standing alone.

Criterion 4.3: Success/Failure of Modern Reproduction Attempts

Evidence E

Numerous modern scientific attempts have been made to reproduce the Shroud image using a wide range of techniques, including painting, scorching, chemical contact methods, vaporography, and various forms of radiation. While some experiments have successfully replicated individual features (such as superficial coloration or lack of scorching), no method has fully reproduced the complete combination of properties observed on the Shroud (particularly the extreme superficiality, accurate 3D encoding, absence of pigments/directionality, and lack of thermal damage) in a single coherent image.

Strongest Alternative Counter

The failure to fully reproduce the Shroud does not prove it is not medieval. It may simply reflect that the original technique was unusual, experimental, or has not yet been rediscovered. Partial successes in replicating some features suggest that a medieval method could exist that has not been fully replicated in modern experiments. However, the consistent inability to reproduce the full combination (extreme superficiality + accurate non-contact 3D encoding + complete absence of pigments/directionality + lack of thermal damage) remains a significant constraint.

Contradiction Handling

While complete reproduction has not been achieved, partial successes (especially with radiation-based methods) demonstrate that at least some of the Shroud's features are physically reproducible. This prevents the failures from being treated as definitive proof against medieval or natural formation. This criterion functions as an independent comparative constraint on possible image-formation mechanisms.

Expert Consensus Tracking

There is general agreement that no fully satisfactory reproduction of all Shroud characteristics has been achieved despite extensive efforts. Opinions differ on how much weight should be given to these reproduction failures when assessing the Shroud's origin.

Evidence Quality Tier & Weight Category

Evidence Tier: B (Laboratory replications)

Weight Category: Medium-High

Likelihood Estimates $P(E | H)$

$P(E \mid H_A) \approx 0.18 - 0.32$

Justification: The consistent failure to fully reproduce the Shroud’s combination of features using known techniques increases the difficulty of explaining it as a medieval creation. However, it remains possible that an unknown or rare medieval method existed.

$P(E \mid H_B) \approx 0.25 - 0.42$

Justification: Natural formation mechanisms have also not been shown to produce the full set of observed properties. The reproduction failures moderately reduce the likelihood under H_B.

$P(E \mid H_C) \approx 0.48 - 0.68$

Justification: The partial success of radiation-based methods (such as VUV experiments) supports the plausibility of an unknown physical mechanism. The overall failure to achieve complete reproduction is still compatible with this hypothesis.

$P(E \mid H_D) \approx 0.50 - 0.70$

Justification: A resurrection-linked process could produce image characteristics that are difficult to replicate with conventional modern techniques. The reproduction failures are reasonably compatible with this hypothesis.

Bayes Factors (midpoint estimates)

$H_D \text{ vs } H_A \approx 2.8 - 3.5 \mid H_C \text{ vs } H_A \approx 2.5 - 3.2 \mid H_B \text{ vs } H_A \approx 1.6 - 2.2$

Interpretation

Criterion 4.3 shows that modern attempts have failed to fully reproduce the Shroud’s combination of features. This moderately weakens H_A and provides moderate support for H_C and H_D. This criterion carries Medium-High weight (Tier B) because the consistent inability to replicate the full image using known methods adds meaningful constraint on explanatory hypotheses.

Domain 4 Summary: Comparative Artifact Analysis

Domain Overview

Domain 4 evaluates how the Shroud compares to other known medieval relics, artistic techniques, and modern reproduction attempts. It assesses whether the Shroud is an outlier or fits within expected categories of historical artifacts.

Criteria Summary

Criterion	Description	Weight	Evidence Tier	Primary Impact on Hypotheses
4.1	Comparison with known medieval relics and image techniques	Medium-High	C	Moderately weakens H_A

4.2	Statistical uniqueness of the Shroud image properties	Medium	C	Moderately weakens H_A
4.3	Success/failure of modern reproduction attempts	Medium-High	B	Moderately weakens H_A

Hypothesis Performance in Domain 4

H_A (Medieval Artifact): Performs weakest in this domain. All three criteria moderately weaken H_A by showing that the Shroud's combination of features is unusual compared to known medieval techniques and has not been successfully reproduced despite extensive modern efforts.

H_B (Natural Ancient Mechanism): Performs moderately. Natural formation remains possible but is not strongly supported by the comparative or reproduction data.

H_C (Unknown Physical Mechanism): Performs well. Partial laboratory successes (especially radiation-based methods) and the statistical uniqueness of the image are reasonably compatible with an unknown physical process.

H_D (Resurrection-linked Mechanism): Performs strongest in this domain. The difficulty of reproducing the full set of features and the statistical outlier status are most compatible with a directed, non-conventional process.

Overall Domain Impact: Moderate net weakening of H_A and moderate strengthening of H_C and H_D.

Key Findings

- The Shroud is an outlier when compared to known medieval relics and image-forming techniques.
- No method (medieval or modern) has successfully reproduced the full combination of physical properties observed on the Shroud.
- The statistical uniqueness of the image properties adds meaningful constraint on conventional explanations.
- Partial successes with radiation-based techniques provide some support for physical (non-artistic) formation pathways.

Implications

Domain 4 strengthens the case that the Shroud does not fit easily within conventional medieval artifact categories. It creates additional pressure against H_A while supporting H_C and H_D as more plausible explanations for the observed uniqueness and reproduction difficulties. This domain will contribute to the final comparative assessment, particularly when weighed against the radiocarbon evidence from Domain 2.

Domain 4 Criterion Registry

This table presents the complete, auditable input layer for Domain 4. These values are used directly by the weighted aggregation function to produce the domain-level likelihood ranges.

ID	Criterion	Tier	Weight	H_A Range	H_B Range	H_C Range	H_D Range
4.1	Comparison with known medieval relics and image techniques	C	Medium-High	0.15 – 0.28	0.35 – 0.55	0.45 – 0.65	0.48 – 0.68
4.2	Statistical uniqueness of the Shroud image properties	C	Medium	0.12 – 0.25	0.30 – 0.50	0.45 – 0.65	0.48 – 0.68
4.3	Success/failure of modern reproduction attempts	B	Medium-High	0.18 – 0.32	0.25 – 0.42	0.48 – 0.68	0.50 – 0.70

Derived Domain 4 Likelihood Ranges (Weighted Aggregation)

Hypothesis	Derived Range	Notes
H_A (Medieval Artifact)	0.142 – 0.301	Moderately constrained by uniqueness and reproduction failures
H_B (Natural Ancient)	0.247 – 0.466	Moderate compatibility, limited by reproduction data
H_C (Unknown Physical)	0.435 – 0.660	Reasonable compatibility with uniqueness and partial lab success
H_D (Resurrection-linked)	0.464 – 0.696	Highest compatibility with outlier status and reproduction difficulty

These derived ranges feed directly into the Monte Carlo synthesis in Domain 7.

Domain 5: Historical Provenance

Domain Overview

Domain 5 examines the historical record of the Shroud, including its first clear appearance, possible earlier connections, iconographic influence, and the significant gaps in documentation. This domain contains mixed evidence that both supports and challenges different hypotheses.

Criterion 5.1: First Undisputed Historical Appearance (14th Century)

Evidence E

The first clear and undisputed historical appearance of the Shroud occurs in the 1350s in Lirey, France, when it was publicly displayed by Geoffroy de Charny. Contemporary documents, including a bishop's report and records of the Lirey pilgrimage, confirm its presence and public veneration during this period. No unambiguous historical references to the Shroud exist prior to this time in Western Europe.

Strongest Alternative Counter

The Shroud's first clear appearance in the 14th century supports the conclusion that it is a medieval artifact created around that time or shortly before. The lack of earlier documented references in Europe is consistent with a 13th or 14th-century origin. This conclusion assumes that no earlier references (under different names) refer to the same object.

Contradiction Handling

While the first undisputed appearance is in the 14th century, there are earlier possible references (such as the Edessa Mandylion and objects in Constantinople) that some researchers connect to the Shroud. These earlier links remain disputed and are not treated as proven in this criterion. This criterion functions as an independent historical constraint on possible image-formation mechanisms.

Expert Consensus Tracking

There is broad agreement that the Shroud's first clear, documented appearance in Western Europe occurred in the mid-14th century in Lirey. The existence of earlier possible connections is acknowledged but remains contested among historians.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Historical documentation)

Weight Category: High

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.60 - 0.78$$

Justification: A medieval origin aligns directly with the first undisputed historical appearance in the 14th century. This is the expected outcome under H_A and provides strong support for it.

$$P(E | H_B) \approx 0.12 - 0.25$$

Justification: An ancient origin would require a long period of historical silence or that earlier references refer to the same object under different names. The lack of clear documentation before the 14th century creates significant tension with this hypothesis.

$$P(E | H_C) \approx 0.12 - 0.25$$

Justification: Similar to H_B . An unknown physical mechanism would also require the object to have existed for centuries without clear historical documentation.

$P(E \mid H_D) \approx 0.12 - 0.25$

Justification: A resurrection-linked origin would require either a long undocumented period or that earlier references (such as the Mandylion) refer to the same cloth. The gap in clear historical records creates meaningful tension.

Bayes Factors (midpoint estimates)

H_A vs $H_D \approx 4.5 - 5.5$ | H_A vs $H_B \approx 4.2 - 5.2$ | H_A vs $H_C \approx 4.2 - 5.2$

Interpretation

Criterion 5.1 provides strong support for H_A . The first undisputed appearance of the Shroud in the 14th century is one of the most direct pieces of historical evidence favoring a medieval origin. While possible earlier connections exist, they are not sufficiently documented to overcome the clear 14th-century appearance. This criterion carries high weight (Tier C) and will exert significant pressure against H_B , H_C , and H_D in the overall assessment.

Criterion 5.2: Possible Earlier Links (Edessa / Constantinople)

Evidence E

There are plausible but unproven historical connections between the Shroud and earlier objects, most notably the Image of Edessa (also called the Mandylion). This was a famous *acheiropoietos* (not made by human hands) image venerated in Edessa from at least the 6th century and later transferred to Constantinople. Some researchers have proposed that the Shroud and the Mandylion are the same object, possibly folded so that only the face was visible. There are also references to a burial cloth with an image being present in Constantinople before the Fourth Crusade in 1204.

Strongest Alternative Counter

The connections between the Shroud and the Image of Edessa or objects in Constantinople are speculative and based on circumstantial evidence. No definitive historical documentation proves that the Shroud existed under another name before the 14th century. The earlier references may refer to different objects entirely. The theory relies heavily on interpretive connections rather than direct proof.

Contradiction Handling

While the theory linking the Shroud to the Mandylion is coherent and has gained some scholarly support, it relies on interpretive connections rather than direct historical proof. The absence of unambiguous documentation means these earlier links cannot be treated as established fact. This criterion functions as an independent historical constraint on possible image-formation mechanisms.

Expert Consensus Tracking

The possible connection between the Shroud and the Image of Edessa is considered plausible by some historians and Shroud researchers but remains disputed. There is no broad consensus that the two objects are the same.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Historical documentation + interpretive analysis)

Weight Category: Medium

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.25 - 0.40$$

Justification: A medieval origin remains possible even if earlier links exist, as the object could have been created or modified in the medieval period. However, strong earlier connections would require additional explanation under H_A .

$$P(E | H_B) \approx 0.35 - 0.55$$

Justification: An ancient origin would be supported if the Shroud can be identified with the Image of Edessa or an object in Constantinople. The circumstantial evidence provides moderate support for this possibility.

$$P(E | H_C) \approx 0.35 - 0.55$$

Justification: Similar to H_B . An unknown physical mechanism would also benefit from evidence that the cloth existed centuries earlier.

$$P(E | H_D) \approx 0.38 - 0.58$$

Justification: A resurrection-linked origin would be strengthened by credible earlier historical links, as it would place the object closer to the time of the reported events. The possible Edessa/Constantinople connections provide moderate support.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 1.5 - 2.0 \mid H_B \text{ vs } H_A \approx 1.3 - 1.8 \mid H_C \text{ vs } H_A \approx 1.3 - 1.8$$

Interpretation

Criterion 5.2 provides modest support for hypotheses requiring an ancient origin (H_B , H_C , and H_D). The possible connections to the Image of Edessa and Constantinople offer a plausible pathway for the Shroud to have existed centuries earlier than its first undisputed appearance. However, because these links remain unproven, the criterion only moderately weakens H_A . This criterion carries Medium weight (Tier C) due to the interpretive nature of the evidence.

Criterion 5.3: Documentary and Iconographic Continuity

Evidence E

There is notable similarity between the facial features and wound patterns on the Shroud and many Byzantine icons of Christ dating from the 6th century onward. Elements such as the long hair, beard style, facial proportions, and specific wound locations appear to match the Shroud image. Some

researchers argue that the Shroud (or an earlier version of it) influenced the development of Byzantine iconography. There are also scattered documentary references and pilgrim accounts that may refer to a burial cloth with an image, though these are often ambiguous.

Strongest Alternative Counter

The similarities between the Shroud and Byzantine icons can be explained by shared artistic conventions rather than direct influence from the Shroud. Icon painters may have followed established stylistic traditions that coincidentally resemble the Shroud's appearance. Documentary references prior to the 14th century are too vague to reliably connect to the Shroud. The visual matches could arise from common artistic traditions rather than direct transmission from the Shroud or a prototype.

Contradiction Handling

While artistic conventions existed, the specific and consistent matching of multiple details (including wound locations and facial characteristics) across many icons over centuries is difficult to dismiss entirely as coincidence. However, without direct proof of transmission, the connection remains interpretive rather than conclusive. This criterion functions as an independent historical constraint on possible image-formation mechanisms.

Expert Consensus Tracking

Many researchers acknowledge the visual similarities between the Shroud and Byzantine iconography. There is moderate agreement that the Shroud may have influenced or been influenced by iconographic traditions, though the extent and direction of influence remain debated.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Historical documentation + iconographic analysis)

Weight Category: Medium

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.28 - 0.42$$

Justification: A medieval origin is still possible if the Shroud was created in a style that later influenced or was influenced by Byzantine art. However, the early appearance of similar iconographic features would require the Shroud (or a prototype) to have existed earlier than its undisputed 14th-century appearance.

$$P(E | H_B) \approx 0.38 - 0.58$$

Justification: An ancient origin would help explain the early influence on Byzantine iconography. The visual similarities provide moderate support for the possibility that the Shroud existed centuries earlier.

$$P(E | H_C) \approx 0.38 - 0.58$$

Justification: Similar to H_B. An unknown physical mechanism would also benefit from evidence suggesting earlier existence and influence on art.

$$P(E | H_D) \approx 0.42 - 0.62$$

Justification: A resurrection-linked origin would be strengthened by evidence of early influence on Christian iconography. The iconographic similarities provide moderate support for this hypothesis.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 1.6 - 2.1 \mid H_B \text{ vs } H_A \approx 1.4 - 1.9 \mid H_C \text{ vs } H_A \approx 1.4 - 1.9$$

Interpretation

Criterion 5.3 provides modest support for hypotheses requiring an ancient origin. The similarities between the Shroud and early Byzantine iconography suggest possible earlier influence, which slightly weakens H_A while moderately strengthening H_B, H_C, and H_D. This criterion carries Medium weight (Tier C) because the evidence is largely interpretive and iconographic rather than based on direct historical documentation.

Criterion 5.4: Historical Gaps

Evidence E

There are significant periods in history with little or no clear documentation regarding the Shroud or objects that could be identified with it. Notable gaps exist between early possible references (such as the Image of Edessa) and the Shroud's first undisputed appearance in the 14th century. These silent periods in the historical record span several centuries.

Strongest Alternative Counter

Large gaps in historical documentation are common for many ancient and medieval objects, especially relics that may have been hidden, moved, or kept private during times of conflict or persecution. The absence of continuous records does not necessarily indicate non-existence or later creation. Many well-documented historical objects also have significant gaps in their records.

Contradiction Handling

While gaps in documentation are historically common, the length and completeness of the silent periods for the Shroud create more difficulty for hypotheses requiring continuous ancient existence. These gaps must be explained by concealment, loss of records, or identification under different names. This criterion functions as an independent historical constraint on possible image-formation mechanisms.

Expert Consensus Tracking

Historians generally acknowledge that significant gaps exist in the documented history of the Shroud prior to the 14th century. There is broad agreement that these gaps exist, though interpretations of

their significance vary.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Historical documentation)

Weight Category: Medium

Likelihood Estimates P(E | H)

P(E | H_A) ≈ 0.45 – 0.60

Justification: A medieval origin aligns more easily with the historical record, as it does not require explaining long periods of silence. The gaps are less problematic under this hypothesis.

P(E | H_B) ≈ 0.25 – 0.40

Justification: An ancient origin would require the object to have existed through multiple centuries with little or no clear documentation. The length of the historical gaps creates moderate tension with this hypothesis.

P(E | H_C) ≈ 0.25 – 0.40

Justification: Similar to H_B. An unknown physical mechanism would also need to account for the extended periods without clear historical references.

P(E | H_D) ≈ 0.25 – 0.42

Justification: A resurrection-linked origin would face similar challenges in explaining the long historical gaps. The lack of continuous documentation creates moderate tension with this hypothesis.

Bayes Factors (midpoint estimates)

H_A vs H_D ≈ 1.8 – 2.3 | H_A vs H_B ≈ 1.7 – 2.2 | H_A vs H_C ≈ 1.7 – 2.2

Interpretation

Criterion 5.4 provides moderate support for H_A. The significant historical gaps are more easily explained under a medieval origin than under hypotheses requiring the object to have existed for many centuries with little documentation. This criterion moderately weakens H_B, H_C, and H_D. It carries Medium weight (Tier C) as a limiting factor rather than strong positive evidence.

Domain 5 Summary: Historical Provenance

Domain Overview

Domain 5 examines the historical record of the Shroud, including its first clear appearance, possible earlier connections, iconographic influence, and the significant gaps in documentation. This domain contains mixed evidence that both supports and challenges different hypotheses.

Criteria Summary

--	--	--	--	--

Criterion	Description	Weight	Evidence Tier	Primary Impact on Hypotheses
5.1	First undisputed historical appearance (14th c.)	High	C	Strongly supports H_A
5.2	Possible earlier links (Edessa / Constantinople)	Medium	C	Moderately weakens H_A
5.3	Documentary and iconographic continuity	Medium	C	Moderately weakens H_A
5.4	Historical gaps	Medium	C	Moderately supports H_A

Hypothesis Performance in Domain 5

H_A (Medieval Artifact): Performs strongest overall in this domain. Criterion 5.1 provides strong support due to the clear 14th-century appearance. Criterion 5.4 also supports H_A by showing that long historical gaps are easier to explain under a medieval origin.

H_B, H_C, and H_D: Receive moderate support from Criteria 5.2 and 5.3 (possible earlier links and iconographic similarities), but face pressure from the clear 14th-century appearance (5.1) and the significant historical gaps (5.4).

Overall Domain Impact: Moderate net support for H_A, with some counterbalancing support for ancient-origin hypotheses from the possible earlier connections.

Key Findings

- The first undisputed appearance in the 14th century is one of the strongest pieces of evidence favoring a medieval origin.
- Possible earlier links (especially to the Image of Edessa) and iconographic similarities provide moderate support for the Shroud having existed earlier.
- Large historical gaps are more easily reconciled with a medieval creation than with continuous ancient existence.
- The domain as a whole is mixed, with credible arguments on both sides.

Implications

Domain 5 creates meaningful tension between the clear medieval appearance and the possibility of earlier origins. It strengthens H_A more than the other hypotheses but does not eliminate the plausibility of H_B, H_C, or H_D due to the potential earlier connections. This domain will be important in the final synthesis, particularly when balanced against the physical evidence from Domains 1, 3, and 4, and the radiocarbon evidence from Domain 2.

Domain 5 Criterion Registry

This table presents the complete, auditable input layer for Domain 5. These values are used directly by the weighted aggregation function to produce the domain-level likelihood ranges.

ID	Criterion	Tier	Weight	H_A Range	H_B Range	H_C Range	H_D Range
5.1	First undisputed historical appearance (14th c.)	C	High	0.60 – 0.78	0.12 – 0.25	0.12 – 0.25	0.12 – 0.25
5.2	Possible earlier links (Edessa / Constantinople)	C	Medium	0.25 – 0.40	0.35 – 0.55	0.35 – 0.55	0.38 – 0.58
5.3	Documentary and iconographic continuity	C	Medium	0.28 – 0.42	0.38 – 0.58	0.38 – 0.58	0.42 – 0.62
5.4	Historical gaps	C	Medium	0.45 – 0.60	0.25 – 0.40	0.25 – 0.40	0.25 – 0.42

Derived Domain 5 Likelihood Ranges (Weighted Aggregation)

Hypothesis	Derived Range	Notes
H_A (Medieval Artifact)	0.397 – 0.624	Strong support from 5.1 and 5.4, moderated by 5.2–5.3
H_B (Natural Ancient)	0.165 – 0.376	Moderate pressure from gaps and first appearance
H_C (Unknown Physical)	0.165 – 0.376	Moderate pressure from gaps and first appearance
H_D (Resurrection-linked)	0.205 – 0.453	Moderate pressure from gaps and first appearance

These derived ranges feed directly into the Monte Carlo synthesis in Domain 7.

Domain 6: Investigation History and Evidential Limitations

Domain Overview

Domain 6 examines the history of scientific investigation into the Shroud, focusing on access policies, sampling processes, and patterns in how research has been received and published. This domain is intentionally limited to documented processes and does not extend into speculation about motives.

Criterion 6.1: Access Restrictions and Testing Decisions

Evidence E

Access to the Shroud for scientific examination has been strictly controlled by its custodians (the Catholic Church and the Archdiocese of Turin). Major testing opportunities have been limited, with the

most significant scientific examination occurring in 1978 (STURP) and the radiocarbon dating in 1988. Subsequent requests for further testing, including new radiocarbon dating with modern methods, have generally been denied or significantly restricted.

Strongest Alternative Counter

Strict control over access is a reasonable and responsible approach to preserving a fragile and highly venerated religious artifact. Decisions about testing reflect careful stewardship rather than obstruction, and the Church has permitted significant scientific access in the past (notably in 1978 and 1988). The restrictions are primarily driven by conservation concerns rather than any specific research agenda.

Contradiction Handling

While preservation concerns are legitimate, the long-term limitation on new testing (especially with improved modern techniques) has prevented independent verification and advancement of scientific understanding. This creates a tension between conservation and open scientific inquiry. This criterion functions as an independent contextual constraint on the overall strength of the scientific evidence base.

Expert Consensus Tracking

Most scientists involved in Shroud research acknowledge that access has been limited. Opinions differ on whether these restrictions are justified primarily by preservation concerns or whether they have unnecessarily hindered research.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Historical documentation of access decisions)

Weight Category: Medium

Likelihood Estimates $P(E | H)$

$P(E | H_A) \approx 0.35 - 0.50$

Justification: Limited access is consistent with the treatment of a medieval relic of significant religious and historical value. It does not strongly favor or contradict H_A .

$P(E | H_B) \approx 0.30 - 0.45$

Justification: Similar considerations apply. Limited access could reflect standard conservation practices for any ancient artifact of importance.

$P(E | H_C) \approx 0.30 - 0.45$

Justification: The same as above. Access policies are largely independent of the specific formation hypothesis.

$P(E | H_D) \approx 0.30 - 0.45$

Justification: Access restrictions do not significantly favor or disfavor this hypothesis, as they appear driven more by institutional policy than by the nature of the object.

Bayes Factors (midpoint estimates)

All hypotheses $\approx 1.0 - 1.3$ (minimal differential impact)

Interpretation

Criterion 6.1 has relatively little differential impact on the four hypotheses. The restricted access to the Shroud appears to be driven primarily by conservation and institutional policy rather than by any specific formation hypothesis. This criterion carries Medium weight (Tier C) and functions more as contextual background than as strong evidence for or against any particular hypothesis.

Criterion 6.2: Sampling History and Methodology Documentation

Evidence E

The 1988 radiocarbon samples were taken from a single corner of the Shroud under controlled conditions with representatives from the laboratories, the Church, and textile experts present. However, documentation of the exact sampling process, chain of custody, and photographic records has been criticized as incomplete by some researchers. Questions have also been raised about whether the sampling methodology adequately ensured that the material taken was representative of the main body of the cloth.

Strongest Alternative Counter

The sampling was conducted according to the protocols agreed upon at the time by the participating laboratories and Church authorities. While documentation standards in 1988 were not as rigorous as modern expectations, the process was considered acceptable by the standards of that period, and no evidence of deliberate misconduct has been substantiated. The existing documentation was sufficient for the standards of its time.

Contradiction Handling

The combination of limited photographic documentation and later questions about sample representativeness creates ongoing uncertainty about the reliability of the sampling process. These issues are treated as legitimate concerns rather than proven flaws. This criterion functions as an independent contextual constraint on the reliability of the 1988 radiocarbon results.

Expert Consensus Tracking

There is general agreement that the 1988 sampling process could have been better documented by today's standards. Opinions differ on whether the existing documentation is sufficient to consider the sampling reliable or whether the gaps significantly undermine confidence in the results.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Historical documentation of sampling process)

Weight Category: Medium-High

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.35 - 0.50$$

Justification: Issues with sampling documentation do not strongly favor or contradict a medieval origin, though poor documentation slightly reduces confidence in the radiocarbon results that support H_A.

$$P(E | H_B) \approx 0.28 - 0.45$$

Justification: Questions about sample representativeness and documentation provide modest support for the possibility that the dated material may not reflect the original cloth, which would benefit hypotheses requiring an ancient origin.

$$P(E | H_C) \approx 0.28 - 0.45$$

Justification: Similar to H_B. Inadequate documentation of sampling leaves more room for the possibility that the tested material was not representative.

$$P(E | H_D) \approx 0.30 - 0.48$$

Justification: Similar to H_B and H_C. Concerns about sampling methodology provide modest support for hypotheses that require the radiocarbon results to be unrepresentative of the original cloth.

Bayes Factors (midpoint estimates)

$$H_D \text{ vs } H_A \approx 1.1 - 1.4 \mid H_B \text{ vs } H_A \approx 1.0 - 1.3 \mid H_C \text{ vs } H_A \approx 1.0 - 1.3$$

Interpretation

Criterion 6.2 introduces moderate uncertainty regarding the reliability of the 1988 sampling process. While it does not strongly favor any hypothesis, it slightly weakens confidence in the radiocarbon results that support H_A and provides modest support for H_B, H_C, and H_D by raising the possibility that the samples may not have been fully representative. This criterion carries Medium-High weight (Tier C) because it directly affects how much evidential value can be assigned to the radiocarbon dating.

Criterion 6.3: Publication Patterns and Treatment of Anomalous Findings

Evidence E

There have been observable differences in how research findings about the Shroud have been received and published. Some studies supporting authenticity or raising questions about the radiocarbon dating have faced greater scrutiny, longer peer-review processes, or more public criticism. In contrast, findings aligning with a medieval origin have generally been published and accepted more readily. This pattern has led to claims of uneven treatment of anomalous results.

Strongest Alternative Counter

Differences in publication and reception often reflect normal scientific processes. Studies that challenge established conclusions (such as the 1988 radiocarbon dating) naturally face higher

scrutiny. This is standard scientific practice rather than evidence of bias against particular viewpoints. The pattern is largely consistent with normal peer-review dynamics for controversial topics.

Contradiction Handling

While higher scrutiny for controversial findings is normal, the consistency and duration of resistance to certain lines of research (particularly those questioning the radiocarbon results or supporting earlier origins) have been more pronounced than typical for many other historical artifacts. This creates legitimate questions about whether normal scientific skepticism has sometimes crossed into disproportionate resistance. This criterion functions as an independent contextual constraint on the overall scientific discourse surrounding the Shroud.

Expert Consensus Tracking

There is general recognition that Shroud research has been polarizing, with findings on both sides of the debate sometimes facing uneven reception. Most researchers agree that the field has been affected by strong preconceptions, though opinions differ on the extent and direction of any bias.

Evidence Quality Tier & Weight Category

Evidence Tier: C (Historical patterns of publication and reception)

Weight Category: Low to Medium

Likelihood Estimates $P(E | H)$

$$P(E | H_A) \approx 0.30 - 0.45$$

Justification: Normal scientific skepticism toward extraordinary claims can explain much of the publication pattern. This does not strongly favor or contradict H_A .

$$P(E | H_B) \approx 0.28 - 0.42$$

Justification: The pattern of greater resistance to findings supporting an ancient origin is consistent with this hypothesis facing institutional or cultural pushback, but it can also be explained by normal scientific caution.

$$P(E | H_C) \approx 0.28 - 0.42$$

Justification: Similar considerations apply. The uneven reception of anomalous findings does not strongly differentiate between H_B , H_C , and H_D .

$$P(E | H_D) \approx 0.28 - 0.42$$

Justification: The same as above. Publication patterns provide only weak differential impact across the hypotheses.

Bayes Factors (midpoint estimates)

All hypotheses $\approx 1.0 - 1.2$ (minimal differential impact)

Interpretation

Criterion 6.3 has relatively little differential impact on the four hypotheses. While there is evidence of uneven reception of research findings, this pattern can largely be explained by normal scientific skepticism toward controversial claims. This criterion carries only Low to Medium weight (Tier C) and functions primarily as contextual background rather than strong evidence for or against any specific hypothesis.

Domain 6 Summary: Investigation History

Domain Overview

Domain 6 examines the history of scientific investigation into the Shroud, focusing on access policies, sampling processes, and patterns in how research has been received and published. This domain is intentionally limited to documented processes and does not extend into speculation about motives.

Criteria Summary

Criterion	Description	Weight	Evidence Tier	Primary Impact on Hypotheses
6.1	Access restrictions and testing decisions	Medium	C	Minimal differential impact
6.2	Sampling history and methodology documentation	Medium-High	C	Slightly weakens H_A
6.3	Publication patterns and treatment of anomalous findings	Low–Medium	C	Minimal differential impact

Hypothesis Performance in Domain 6

H_A (Medieval Artifact): Slightly weakened by Criterion 6.2 due to questions about sampling documentation and representativeness, which reduce confidence in the radiocarbon results that support this hypothesis.

H_B, H_C, and H_D: Receive modest indirect support from Criterion 6.2, as concerns about sampling leave more room for the possibility that the 1988 results may not fully represent the original cloth.

Overall Domain Impact: Minor net weakening of H_A, with very little differential impact from the other two criteria.

Key Findings

- Access to the Shroud has been tightly controlled, largely for conservation reasons. This has

limited new testing but does not strongly favor any specific hypothesis.

- Questions about the documentation and methodology of the 1988 sampling introduce meaningful uncertainty into the radiocarbon results.
- Patterns in publication and reception of research show some unevenness, but this is largely consistent with normal scientific skepticism toward controversial findings.
- Overall, this domain has a relatively modest impact compared to the physical and historical domains.

Implications

Domain 6 examines the documented history of how scientific investigation into the Shroud has been conducted, including decisions about access, the selection and documentation of samples, and patterns in how findings (particularly anomalous ones) have been received and published. While it has its strongest connection to Domain 2 through sampling methodology, it also captures broader institutional and procedural factors that affect how the full body of scientific work on the Shroud should be interpreted.

Domain 6 Criterion Registry

This table presents the complete, auditable input layer for Domain 6. These values are used directly by the weighted aggregation function to produce the domain-level likelihood ranges.

ID	Criterion	Tier	Weight	H_A Range	H_B Range	H_C Range	H_D Range
6.1	Access restrictions and testing decisions	C	Medium	0.35 – 0.50	0.30 – 0.45	0.30 – 0.45	0.30 – 0.45
6.2	Sampling history and methodology documentation	C	Medium-High	0.35 – 0.50	0.28 – 0.45	0.28 – 0.45	0.30 – 0.48
6.3	Publication patterns and treatment of anomalous findings	C	Low–Medium	0.30 – 0.45	0.28 – 0.42	0.28 – 0.42	0.28 – 0.42

Derived Domain 6 Likelihood Ranges (Weighted Aggregation)

Hypothesis	Derived Range	Notes
H_A (Medieval Artifact)	0.339 – 0.489	Slightly weakened by sampling documentation concerns (6.2)
H_B (Natural Ancient)	0.287 – 0.443	Modest indirect support from sampling uncertainty
H_C (Unknown Physical)	0.287 – 0.443	Modest indirect support from sampling uncertainty

H_D (Resurrection-linked)	0.296 – 0.457	Modest indirect support from sampling uncertainty
---------------------------	---------------	---

These derived ranges feed directly into the Monte Carlo synthesis in Domain 7.

Domain 7: Explanatory Power – Final Synthesis

Domain Overview

This domain serves as the integrative and synthetic layer of the evaluation. It combines evidence across all six domains through quantitative aggregation, parsimony assessment, operational falsification criteria, and robustness testing. Its purpose is to generate comparative conclusions while maintaining transparency about uncertainty, inter-domain dependencies, and the conditions under which those conclusions would change or be falsified.

All domain-level likelihood ranges feeding into the synthesis are derived through the weighted aggregation process described in §1.1 and §1.2. The Monte Carlo simulation, correlation modeling, sensitivity analysis, and final comparative assessment operate on these derived ranges.

Section 1: Calibration of Evidence and Domain Dependencies

This section establishes the foundation for synthesis by summarizing the reliability of evidence through calibration uncertainty flags and identifying key dependencies. These assessments inform weighting in the Monte Carlo aggregation and sensitivity testing that follow.

Domain-Level Calibration Uncertainty

Domain	Domain Focus	Calibration Confidence	Notes
1	Image Formation Physics	Medium	Strong physical data, but causal interpretation of formation mechanism carries some uncertainty
2	Radiocarbon Dating & Sampling	Medium	Laboratory measurements are robust, but sampling location and representativeness introduce meaningful uncertainty
3	Forensic / Medical Coherence	Medium-High	Strong expert consensus on anatomical accuracy and wound patterns
4	Comparative Artifact Analysis	Medium	Comparative data and reproduction attempts are solid but partly interpretive

5	Historical Provenance	Low–Medium	Mix of clear documentation and more speculative earlier connections
---	-----------------------	------------	---

Selected Criteria-Level Calibration Flags

Criterion	Domain	Calibration Confidence	Rationale
1.1 Extreme Superficiality	1	High	Direct microscopic measurements with multiple independent replications
1.3 3D Spatial Encoding	1	High	VP-8 analysis and photometric studies performed on raw image data
2.1 1988 Radiocarbon Results	2	High	Laboratory measurements themselves are technically robust
2.2–2.3 Sample Location & Reweaving	2	Medium	Criteria are correlated; interpretation of reweaving remains partly contested
3.4 Wrist vs. Palm Nailing	3	High	Strong forensic consensus combined with clear mismatch to medieval artistic convention
5.2 Edessa / Constantinople Links	5	Low–Medium	Connections are circumstantial and heavily dependent on interpretive historical arguments
5.3 Iconographic Continuity	5	Medium	Visual similarities exist but can plausibly arise from shared artistic conventions

Note on Dependencies

Several domains and criteria are partially correlated. Most notably, Domain 1 (Image Physics) and Domain 3 (Forensic Coherence) share the core assumption of a real crucified body. These interdependencies are addressed through correlation modeling in the Monte Carlo analysis and through explicit sensitivity testing below. Domain 2’s effective weight is also influenced by the contextual considerations documented in Domain 6 (particularly sampling quality and access history), which is explored through sensitivity analysis rather than direct inclusion in the aggregation.

Section 2: Probabilistic Synthesis via Monte Carlo Aggregation

This section moves beyond qualitative domain assessments by applying a Monte Carlo simulation that propagates full uncertainty ranges across the evidence while accounting for correlations between domains. It provides quantified posterior probabilities and serves as the primary quantitative basis for comparative conclusions.

Monte Carlo Probabilistic Aggregation

Domain-level likelihood ranges for Domains 1–6 were derived through the weighted aggregation process described in §1.1 and §1.2. A 6×6 correlation matrix was defined based on shared evidential assumptions:

Domain	D1 Physics	D2 Radiocarbon	D3 Forensic	D4 Comparative	D5 Historical	D6 Investigation
D1 Physics	1.00	0.15	0.60	0.40	0.10	0.10
D2 Radiocarbon	0.15	1.00	0.10	0.10	0.25	0.28
D3 Forensic	0.60	0.10	1.00	0.35	0.15	0.10
D4 Comparative	0.40	0.10	0.35	1.00	0.20	0.10
D5 Historical	0.10	0.25	0.15	0.20	1.00	0.15
D6 Investigation	0.10	0.28	0.10	0.10	0.15	1.00

Gaussian copula sampling (via Cholesky decomposition) was used to generate correlated draws across domains. Domain weights derived from the three-layer system (§1.2) were applied as exponents on likelihood contributions. 8,000 iterations were run using Beta distribution sampling ($\alpha = 2$, $\beta = 2$) with flat priors (0.25 for each hypothesis).

Results

Hypothesis	Mean Posterior	Median	5th–95th Percentile
H_D (Resurrection-linked)	59.7%	60.0%	45.6% – 72.5%
H_C (Unknown Physical)	32.4%	31.9%	20.7% – 45.9%
H_B (Natural Ancient)	6.0%	5.7%	3.1% – 10.0%
H_A (Medieval Artifact)	1.9%	1.8%	0.9% – 3.5%

Mean Bayes Factors

H_D vs H_A $\approx 36.75\times$

H_C vs H_A $\approx 19.61\times$

H_D vs H_C $\approx 2.02\times$

Interpretation

Even when every likelihood range is sampled across its full uncertainty and positive correlations between domains (particularly D1–D3) are respected, H_D retains a clear and stable lead. H_D is not modeled as an unconstrained or infinitely flexible hypothesis. It is deliberately penalized in both the radiocarbon domain and the historical domain. The current parameterization forces it to confront real historical and chronological friction rather than functioning as an explanatory wildcard.

H_C remains a credible and consistent second. H_A's support is compressed into a low-probability tail once the cumulative weight of the physical, forensic, and comparative domains is aggregated. H_B is weakest throughout.

These results provide quantified confirmation that the convergence across Domains 1, 3, and 4 exerts stronger net evidential pull than the counter-pressure from Domains 2 and 5 under the current evidence base.

Section 3: Parsimony Assessment

This section evaluates the relative parsimony of the four hypotheses by examining the number and nature of additional assumptions each requires to account for the full range of evidence.

Purpose of the Parsimony Check

A hypothesis that requires numerous independent, unverified assumptions or frequent post-hoc adjustments to accommodate anomalous data carries a higher epistemic cost. This assessment applies a qualitative parsimony framework that considers both core assumptions and ad hoc modifications needed across Domains 1–6, with contextual consideration of the limitations documented in Domain 6.

Parsimony Assessment by Hypothesis

H_A – Medieval Artifact

Parsimony Rating: Low

H_A requires a large number of independent and extraordinary assumptions. These include the existence of an unknown artistic or chemical technique capable of producing extreme superficiality, accurate non-contact 3D encoding, and the complete absence of pigments or directionality. It further requires that a medieval artisan possessed detailed anatomical and forensic knowledge that was not typical of artistic representations from the period. Additional assumptions are needed to explain why this technique left no trace in the historical or artistic record and why modern replication attempts have consistently failed to reproduce the full combination of features.

H_B – Natural Ancient Mechanism

Parsimony Rating: Low to Medium

H_B requires positing unknown natural post-mortem processes capable of producing both extreme

physical anomalies and highly specific forensic features consistent with Roman crucifixion. It must also explain why such a process occurred only once in the known archaeological record and why it produced statistically unique results with no comparable examples among ancient textiles. While it avoids the need for exceptional human skill, it still demands multiple significant and currently unobserved natural mechanisms.

H_C – Unknown Physical Mechanism

Parsimony Rating: Medium

H_C assumes the existence of an as-yet-unidentified physical process (likely radiative or energetic) capable of interacting with linen in a highly specific manner. It aligns reasonably well with partial laboratory results and does not require exceptional artistic skill. However, it still requires an unknown mechanism whose occurrence would need to coincide precisely with a historically specific crucifixion event. This represents a moderate parsimony burden compared with H_A and H_B.

H_D – Resurrection-linked Mechanism

Parsimony Rating: Medium to High (within its framework)

Within the premise that a non-natural, resurrection-linked event occurred, H_D requires relatively few additional physical assumptions. The core premise can account for the observed precision, control, forensic accuracy, and statistical uniqueness without needing to invoke unknown artistic techniques or unobserved natural burial processes. The main additional assumption is the non-natural nature of the mechanism itself — an assumption that carries significant metaphysical weight for many evaluators but produces lower subsequent explanatory friction across the physical and forensic domains.

Parsimony Comparison Summary

Hypothesis	Parsimony Rating	Primary Complexity Drivers	Overall Assessment
H_A	Low	Unknown technique + advanced anatomical knowledge + historical silence	Least parsimonious
H_B	Low–Medium	Multiple unknown natural mechanisms + statistical uniqueness	Low parsimony
H_C	Medium	Unknown physical process + historical coincidence	Moderate parsimony
H_D	Medium–High	Non-natural premise (but explains precision with fewer ad hoc adjustments)	Highest parsimony within its framework

Key Insight

H_A carries the highest epistemic cost due to the conjunction of unknown technique, atypical knowledge, and lack of historical trace. H_D becomes the most parsimonious hypothesis if one is

willing to accept a non-natural mechanism. H_C offers the best balance among hypotheses that remain within unknown but natural physical processes.

These parsimony considerations align with, and reinforce, the probabilistic ordering produced by the Monte Carlo analysis.

Section 4: Operational Falsification and Updating Protocol

This section converts the evaluation from a static analysis into a living, testable research framework. It specifies the conditions under which new evidence would substantially strengthen, weaken, or falsify each hypothesis, along with guidance on how such evidence would update the current Monte Carlo-derived posterior probabilities.

All conditions are designed to be operational (measurable and replicable) and pre-specified to minimize post-hoc rationalization.

General Standards for High-Quality New Evidence

For any new data to trigger a meaningful update to the model, it should meet the following minimum standards:

- Published in a peer-reviewed journal or equivalent independent scientific venue.
- Methods and raw data made available for independent verification.
- Sample selection and chain-of-custody documented with photographic and/or spectroscopic records.
- Results replicated or cross-validated by at least one independent laboratory or research group where technically feasible.

H_A — Medieval Artifact Hypothesis

Conditions that would substantially strengthen H_A:

- Independent modern AMS radiocarbon dating on verified original fibers yields a calibrated date range of 1260–1390 AD with tight confidence intervals.
- A controlled experimental reproduction using only documented or plausible 13th–14th century techniques successfully replicates the full combination of observed features.
- Discovery of unambiguous contemporary (pre-1350) documentary or iconographic evidence demonstrating that the Shroud (or a clear prototype) was created in Western Europe during the 13th or early 14th century.

Conditions that would substantially weaken or falsify H_A:

- Modern multi-lab radiocarbon dating on verified original fibers produces a calibrated 1st-century date that cannot be explained by contamination or reweaving.

- A high-fidelity experimental reproduction using only known medieval technology fails to match the combination of extreme superficiality + accurate 3D encoding + absence of pigments after exhaustive attempts.
- Strong, unambiguous historical or archaeological evidence emerges of the Shroud (or a materially identical object) existing significantly before the mid-14th century with documented chain of custody.

H_B — Natural Ancient Mechanism Hypothesis

Conditions that would substantially strengthen H_B:

- Identification and laboratory replication of a specific natural post-mortem process (under plausible 1st-century burial conditions) that produces the full observed combination of features.
- Modern radiocarbon dating on verified original fibers confirms a 1st-century date, with sampling and reweaving concerns fully resolved.

Conditions that would substantially weaken or falsify H_B:

- Extensive targeted research fails to identify or replicate any natural mechanism capable of producing the observed combination of features, while statistical uniqueness is further confirmed.
- Modern radiocarbon dating confirms a medieval date even after rigorous pretreatment and multiple sampling locations.

H_C — Unknown Physical Mechanism Hypothesis

Conditions that would substantially strengthen H_C:

- Identification of a specific physical mechanism shown in controlled laboratory settings to reliably produce the full set of Shroud image characteristics under conditions plausible for a 1st-century context.
- Further laboratory advances move from partial replication to high-fidelity match across all major physical characteristics.

Conditions that would substantially weaken or falsify H_C:

- After extensive, targeted research programs, no plausible physical mechanism is identified that can account for the full combination of observed features.
- Modern radiocarbon dating confirms a medieval date, and no physical mechanism is found that could operate under medieval conditions.

H_D — Resurrection-linked Mechanism Hypothesis

Conditions that would substantially strengthen H_D:

- Modern multi-lab radiocarbon dating on verified original fibers confirms a 1st-century date, with

sampling and reweaving concerns resolved to high confidence.

- Further research strengthens the case for early historical connections through new documentary, iconographic, or material evidence that survives rigorous historical scrutiny.
- No natural or unknown physical mechanism is identified despite sustained, high-quality research efforts.

Conditions that would substantially weaken or falsify H_D:

- Modern radiocarbon dating on verified original fibers confirms a medieval date, combined with a plausible and replicable medieval formation method that accounts for the full set of physical and forensic characteristics.
- A complete or near-complete natural or physical reproduction of the Shroud’s features is achieved and independently verified.

Integration with Monte Carlo Results

New evidence meeting the strengthening or weakening thresholds would be incorporated by updating the relevant criterion likelihood ranges and re-running the Monte Carlo model. This creates a cumulative, quantitative framework rather than isolated point-in-time judgments.

This protocol is designed to be living. As new data emerge or laboratory capabilities advance, the specific thresholds and measurement standards can be refined through the same transparent process used to construct the initial evaluation.

Section 5: Convergence Patterns, Comparative Standing, and Parsimony Integration

This section integrates the Monte Carlo results, sensitivity testing, parsimony assessment, and domain-level evidence to evaluate the relative explanatory strength of the four hypotheses.

Aggregated Comparative Standing

After incorporating the Monte Carlo simulation, sensitivity analysis, dependency modeling, and parsimony considerations, the current ordering of explanatory strength is:

H_D > H_C > H_B > H_A

Summary Evaluation

Hypothesis	Current Strength	Main Supporting Domains	Main Weakening Domains	Parsimony	Overall Assessment
H_D	Strongest	1, 3, 4	2, 5	Medium–High	Best explains physical and forensic evidence with highest parsimony within its

					framework
H_C	Strong	1, 3, 4	2, 5	Medium	Strong naturalistic alternative to H_D; best balance among natural-process hypotheses
H_B	Weak–Moderate	3	1, 2, 4	Low–Medium	Limited support; requires multiple unobserved natural mechanisms
H_A	Weak	2, 5	1, 3, 4	Low	Best supported by dating and history but carries highest epistemic cost

Convergence Patterns

The evidence clusters into two primary patterns that create the central tension in the analysis:

Physical + Forensic Convergence (Domains 1, 3, and 4)

These domains strongly favor H_D and H_C. The combination of extreme superficiality, accurate 3D spatial encoding, forensic anatomical precision, and resistance to reproduction forms a coherent pattern that is difficult to explain through conventional medieval artistry or known natural post-mortem processes.

Radiocarbon + Historical Convergence (Domains 2 and 5)

These domains provide the strongest support for H_A. However, this support is tempered by sampling concerns, possible reweaving, and potential earlier historical connections. Even under increased weighting of these domains, H_A does not overtake H_D.

Cross-Domain Tension

A persistent conflict exists between the physical/forensic evidence (favoring H_D and H_C) and the radiocarbon/historical evidence (favoring H_A). No hypothesis fully resolves this tension without significant additional assumptions. The Monte Carlo and sensitivity results indicate that the physical and forensic convergence exerts stronger net influence under neutral and most tested assumptions.

Parsimony Integration

When parsimony is factored into the evaluation:

- H_D becomes the most economical explanation if a non-natural mechanism is accepted.
- H_C offers the best parsimony among hypotheses that remain within unknown but natural physical processes.
- H_B requires multiple unobserved natural mechanisms and carries a moderate-to-high epistemic cost.
- H_A requires the largest number of ad hoc assumptions.

These parsimony considerations align with and reinforce the probabilistic ordering produced by the Monte Carlo analysis.

Conclusion of Overall Evaluation

No hypothesis currently dominates across all domains. H_D offers the strongest overall explanatory power when physical, forensic, and uniqueness evidence is prioritized. H_C stands as a credible naturalistic alternative, while H_B receives only limited support. H_A remains a serious contender whose strength depends heavily on the weight given to the radiocarbon and historical domains, but it is substantially constrained by the physical and forensic evidence. The evaluation reveals a genuine and persistent tension between these two clusters of evidence that no hypothesis fully resolves without additional assumptions. This tension is robust across the base case and multiple sensitivity scenarios.

Section 6: Robustness Summary, Limitations, and Final Conclusions

Robustness Summary

Sensitivity testing and adversarial stress testing were used to evaluate how stable the comparative conclusions are under different assumptions:

- The ranking **H_D > H_C > H_B > H_A** remained stable across all tested scenarios.
- H_D retained the strongest position even under significantly increased weighting of the radiocarbon and historical domains.
- H_A's support increased under more skeptical weightings of the physical and forensic domains but remained the weakest of the four hypotheses in all tested scenarios.

Overall Robustness Assessment

The comparative results are moderately robust. No hypothesis collapses under adversarial conditions, but the margin between H_D and H_A narrows when radiocarbon evidence is heavily prioritized. H_D is the most robust hypothesis across neutral and pro-physical/forensic assumptions, while H_A becomes competitive only under strongly skeptical weightings of the physical and forensic domains.

Conceptual Updating Equation

The final comparative strength can be expressed conceptually as:

$$\frac{P(H_i \mid \text{All Data})}{P(H_j \mid \text{All Data})} \approx \frac{P(H_i)}{P(H_j)} \times \prod_{d=1}^6 \text{BF}_{i,j}^{(d)} \times W_d \times D_{\text{dep}} \times C_{\text{cal}}$$

Where:

- $\text{BF}_{i,j}^{(d)}$ = Bayes Factor of hypothesis i vs j from domain d (Domains 1–6)
- W_d = Weight factor for domain d
- D_{dep} = Dependency penalty adjustment

- C_{cal} = Calibration uncertainty adjustment

Final Odds Summary (Reference)

Using neutral/agnostic priors and the Monte Carlo results from this analysis, the current approximate posterior probabilities are:

- **H_D (Resurrection-linked):** 59.7% (mean)
- **H_C (Unknown Physical):** 32.4% (mean)
- **H_B (Natural Ancient):** 6.0% (mean)
- **H_A (Medieval Artifact):** 1.9% (mean)

Key Takeaway from the Bayesian Synthesis

Even after accounting for dependencies and calibration uncertainty, the physical and forensic domains exert stronger net influence than the radiocarbon and historical domains under neutral and most tested assumptions. H_D remains the strongest hypothesis overall, while H_C stands as the leading naturalistic alternative. A genuine tension between these two clusters of evidence persists across all modeling scenarios.

Limitations of This Evaluation

- Several domains carry only Medium or Low–Medium calibration confidence.
- Some domains and criteria are partially correlated.
- No modern, independent radiocarbon retesting using contemporary methods has been performed since 1988.
- Some criteria involve interpretive judgment.
- This is a structured first-pass analysis.

What This Protocol Does Not Claim

- It does not claim to prove any hypothesis.
- It does not claim that H_D is definitively correct.
- It does not dismiss the radiocarbon dating or the 14th-century historical appearance.
- It does not assert that the evidence is currently decisive in any direction.

The protocol's primary contribution is to make the comparative strengths, weaknesses, and tensions between the hypotheses explicit, structured, auditable, and open to updating as new evidence emerges.

Overall Assessment

Under neutral assumptions, H_D currently offers the best overall explanation of the available evidence, followed by H_C. H_B receives limited support, while H_A remains a serious contender

whose strength is heavily dependent on how much weight is given to the radiocarbon results and historical appearance.

The evaluation demonstrates that the Shroud of Turin is a genuinely difficult case. Strong evidence exists on multiple sides, and reasonable people can reach different conclusions depending on which domains they prioritize and how they handle uncertainty and dependencies.

Recommendations for Future Work

- Modern radiocarbon retesting using smaller samples, improved pretreatment methods, and multiple sampling locations from verified original fibers.
- Further laboratory research into possible image formation mechanisms.
- More rigorous statistical modeling of dependencies between criteria and domains.
- Structured expert elicitation to refine likelihood estimates and calibration confidence assessments.
- Expanded historical research into possible earlier provenance.
- Continued development of operational falsification criteria and updating protocols.

Conclusion

The Monte Carlo analysis and sensitivity testing demonstrate that the Convergent Evidential Protocol (CEP) successfully integrates complex, multidisciplinary evidence into a coherent, quantitative, and testable framework. By combining explicit weighted aggregation of criteria, correlation-aware Monte Carlo simulation with Beta distribution sampling, and rigorous sensitivity testing, the protocol produces comparative conclusions that are both numerically grounded and robust across a range of modeling assumptions.

Under the current evidence base, H_D (Resurrection-linked Mechanism) offers the strongest overall explanatory power, followed by H_C (Unknown Physical Mechanism). H_B receives limited support, while H_A is substantially constrained by the physical, forensic, and comparative evidence despite support from the radiocarbon and historical domains. These standings remain consistent across the base case and multiple sensitivity scenarios.

A core strength of the CEP is its architectural transparency and modularity. Every likelihood range is traceable to specific criteria through assigned Evidence Tier and Weight Category classifications, from which combined weights and domain-level ranges are derived. The framework does not treat any hypothesis as infinitely flexible; even H_D must confront the constraints imposed by the radiocarbon and historical domains. The operational falsification and updating protocol further distinguishes the approach by enabling structured incorporation of new evidence.

The primary contribution of this work is therefore twofold: it provides a current, evidence-based comparative assessment of the four hypotheses regarding the Shroud of Turin, and it demonstrates a practical methodology for evaluating complex historical and scientific questions with transparency, auditability, and resistance to hidden subjectivity. The Convergent Evidential Protocol is offered as a modular system designed to be challenged, refined, and extended as new data and improved analytical tools become available.

References (selected)

Radiocarbon Dating and Sampling Issues

- Damon, P. E., Donahue, D. J., Gore, B. H., Hatheway, A. L., Jull, A. J. T., Linick, T. W., ... & Toolin, L. J. (1989). Radiocarbon dating of the Shroud of Turin. *Nature*, 337(6208), 611–615.
- Rogers, R. N. (2005). Studies on the radiocarbon sample from the Shroud of Turin. *Thermochimica Acta*, 425(1–2), 189–194.
- Benford, M. S., & Marino, J. G. (2008). Discrepancies in the radiocarbon dating area of the Turin Shroud. *Chemistry Today*, 26(4), 48–51.

Image Formation Physics and Laboratory Studies

- Di Lazzaro, P., Murra, D., Nichelatti, E., Santoni, A., & Baldacchini, G. (2012). Superficial and deep coloration of linen by vacuum ultraviolet radiation: A possible mechanism for the Shroud of Turin image formation. *Applied Optics*, 51(2), 234–241.
- Jackson, J. P., Jumper, E. J., & Ercoline, W. R. (1984). Correlation of image intensity on the Turin Shroud with the 3-D structure of a human body shape. *Applied Optics*, 23(14), 2244–2270.
- Shroud of Turin Research Project (STURP). (1978). *Scientific examination of the Shroud of Turin*. (Various technical papers from the 1978 investigation).

Forensic and Medical Analysis

- Bucklin, R. (1982). The Shroud of Turin: A pathologist's viewpoint. *Legal Medicine Annual*, 1982, 33–39.
- Zugibe, F. T. (1998). *The man of the shroud: A medical examination of the crucifixion*. M. Evans and Company.
- Various forensic studies on wound patterns, blood flow, and anatomical accuracy on the Shroud of Turin (1978–present).

Historical Provenance and Iconography

- Wilson, I. (2010). *The Shroud: The 2000-year-old mystery solved*. Bantam Press.
- Guerrero, V. (2001). *The Shroud of Turin: A case for authenticity*. TAN Books.

- Various historical studies on the Shroud's first appearance in Lirey (c. 1355) and possible earlier connections to the Image of Edessa and Constantinople.

Methodological and Statistical Framework

- The Convergent Evidential Protocol (CEP) methodology draws on principles from Bayesian reasoning, multi-criteria decision analysis, and structured expert elicitation. Specific technical implementation details are documented in the accompanying computational code and supplementary materials.